Asymmetric News Shocks

Hayden Klok*, Shaun Bond

UQ Business School
The University of Queensland

December 23, 2024

Abstract

We investigate asymmetric news sentiment effects on the U.S. equities market, by using traditional and machine learning sentiment classifiers on articles from a proprietary news dataset of over 120 million items. Through an array of techniques, we show; negative news dominates results, sentiment effects are greatest when volatility is higher, and market return heteroskedasticity subsumes these effects by up to 50%. Most importantly, we demonstrate shocks, reversals, and contagion between both the financial news markets and the stock market, and document long-lasting negative effects on future aggregate returns. Our results have implications as the demand and market for news, and the social transmission channel, can together intensify downward pressure on financial markets, particularly during times of financial distress.

*Corresponding author: h.klok@uq.edu.au, shaun.bond@uq.edu.au.

1 Introduction

Financial news sentiment and social media driven investment activities has attracted widespread attention in recent years. For example, with the worldwide financial shock of the COVID-19 pandemic in 2020, and the widely publicised speculative short-selling of Gamestop by retail investors in 2021 (Verlaine and Banerji 2021).[1] These, and recent similar events, such as increased popularity of speculative cryptocurrencies, has given rise to renewed focus on the role of investor sentiment and it's potential influence on asset pricing. Notable recent studies have proposed psychological channels through which sentiment may exacerbate mispricing (Daniel et al., 2020), while others provide evidence that return predictability of sentiment is strongest when uncertainty is greatest (Birru and Young, 2022). In addition, new natural language processing and machine learning techniques have enabled better quantification of information from text through more nuanced word classification (Garcia et al., 2023).

This paper adds to the current literature as we investigate the predictive role of sentiment derived from news. Using a large proprietary news database from Reuters, we quantify the positivity and negativity of over 140,000 news articles over a 20 year period. These sentiments are used in a series of regression models from asset pricing literature to investigate sentiments predictive power in forecasting the daily market returns of the S&P 500. Through the various models, different potential channels through which sentiment effects may act are explored, including; dependence on low (high) market volatility state, the presence of return heteroskedasticity, asymmetric shocks from positive and negative news, and long-term effects at lower frequencies.

We demonstrate three key points. First, negative sentiment dominates positive sentiment in regression models, with both larger magnitude and greater statistical significance. Further, while sentiment magnitude increases during times of higher market volatility, predictive power decreases. By accounting for lagged variance terms in the regression process, sentiment significance and magnitude decreases, suggesting sentiment explains some aspect of return volatility, but is partly subsumed when market volatility is taken into account. Second, exogenous shocks to daily news and returns have long lasting effects on each other when volatility is highest, with shocks to news sentiments the primary driver of future news sentiments. Finally, wavelet analysis suggests sentiment

[1] New York Times (2021). *'Dumb Money' Is on GameStop, and It's Beating Wall Street at Its Own Game.* https://www.nytimes.com/2021/01/27/business/gamestop-wall-street-bets.html. Accessed: 1 August 2022.

effects may culminate and act over multiple frequencies simultaneously when markets are most volatile. During times of major financial crises, clustering is observed, as sentiment effects across multiple frequencies align. Together, our results highlight that news and financial equity markets exist as two connected, but separate independent information systems, with information exchanges, shocks and contagion effects observed between them. Our results highlight the information content of news text, that it captures underlying structural price patterns of the aggregate U.S. equity market, and that news shocks can potentially exacerbate existing market volatility, especially during times of greater uncertainty.

The literature on financial sentiment has grown over the years. Founded on theories of behavioural models of biases and mispricing arising from incorrect shared beliefs (De Long et al., 1990; Minsky, 1977), the information content of news and media reports has been associated with future share price movements, earnings, and liquidity. Further, research suggests this association goes beyond correlation and posits a causal link between the information in news reports and financial market pricing, volatility, and trading volumes. Two notable works include the Baker and Wurgler (2006) proxy of investor sentiment, or more accurately market speculation, from stock fundamentals, and Tetlock (2007), who finds a link between the pessimism of a popular Wall Street Journal column and future downward pressure on stock prices. These studies were followed by renewed interest in investor sentiment, with subsequent studies exploring the relationship between investor sentiment and different aspects of finance, including; future earnings (Tetlock et al., 2008), local investor trading (Engelberg and Parsons, 2011), stock trading volume, volatility, earnings, and fraud (Loughran and McDonald, 2011), and social media coverage and future prices and earnings (Chen et al., 2014). In addition, several studies have investigated the information content of different alternative information sources, for example via proxying for investor sentiment via Google searches (Da et al., 2015; Gao et al., 2020), and by classifying sentiment from images in news articles (Obaid and Pukthuanthong, 2022). Finally, the recently presented theoretical models of social transmission by Hirshleifer (2020) highlight different pathways that may give rise to various biases which could exacerbate mispricing, and these frameworks highlight the importance of economic sentiment, particularly when transmitted via mainstream news sources.

Within asset pricing literature, there are somewhat mixed results and views regarding the role of sentiment. Studies on stock market returns that emphasize the importance of sentiment generally identify aggregate pessimistic sentiments forecast future return reversals in-line with behavioural

models of initial investor overreaction that subsequently unwinds, (Garcia, 2013; Birru and Young, 2022; Obaid and Pukthuanthong, 2022). Results from these studies find consistent evidence of momentum patterns consistent with prior literature of proposed collectively shared biases that result in mispricing (De Long et al., 1990; Hirshleifer, 2020). Still, studies they typically stop short of claiming a causal link, as news driven investor sentiment may occur contemporaneously with fundamental information disclosure, and hence it may not be possible to disentangle the purely news sentiment effect.[2]

In contrast, results from other studies suggest news sentiments contribute little or no new information. For example Hendershott et al. (2015) suggests institutions are informed about news prior to release, based on their ex-ante trading behaviour which suggests significant price discovery. Ke et al. (2019) provides evidence that novel news information and sentiment is quickly absorbed into prices within a one day window. Even the work of Tetlock (2007) finds mixed results, with lagged sentiments predicting next day returns, but also lagged returns predicting next day sentiments, with both being statistically significant. Though financial news coverage, and by extension financial sentiment, may help to reduce asymmetry of information and lead to more efficient price discovery (Tetlock, 2010), other studies show that sentiment can disrupt the risk-return trade-off during times of high uncertainty (Yu and Yuan, 2011). Mixed news signals may increase differences in investors' opinions, and may increase limits to arbitrage, particularly during times of greater uncertainty when firms may be harder to value and a wider range of beliefs around fundamentals exist. During these times investors may be more susceptible to irrational price views, and may be more susceptible to their own biases (Zhang, 2006). Further, arbitrageurs and rational investors, who would otherwise help facilitate price-corrections may be discouraged to participate in the markets due to the risks or costs involved, and asset managers may be incentivised to ride bubbles driven by collective investor sentiments (Abreu and Brunnermeier, 2003). In such situations, feedback effects within news reporting could exacerbate price deviations from fundamentals, and could contribute to large price corrections or crashes.

Given the somewhat mixed results in the literature, this paper aims to clarify the role of news sentiments in forecasting future market returns. News sentiments may be driven by changes in returns and vice versa, but may also help to forecast future returns and return volatility. News

[2]One exception is Engelberg and Parsons (2011) as they compare investor behaviours with different levels of news coverage, however in this case one must take into account attention effects, see for example (Da et al., 2011). Hence sentiment effects may be stronger if limited information is available, and attention is more focused on a particular asset or information source.

sentiment may exacerbate short-term return anomalies due to collective investor overreaction, and may have lasting effects particularly during times of greater uncertainty.

News reports and alerts are one of the most common sources used for proxying investor sentiments, and various approaches towards sentiment classification of text exist. The earliest Natural Language Processing (NLP) techniques were lexicon based, simple bag-of-word dictionary approaches, such as Loughran and McDonald (2011). These were followed by more sophisticated rule-based techniques and weighted dictionaries, such as VADER (Hutto and Gilbert, 2014). More recently, machine learning advancements have given rise to new types of algorithms, including transformer neural networks, which are well-suited to various text processing tasks (Vaswani, 2017). One of the most widely-recognised transformers is BERT (Devlin, 2018), which has been demonstrated as a particularly versatile model, the success of which has given rise to various subsequent models. Two models particularly relevant to sentiment analysis and finance are FinBERT (Araci, 2019) and TwitterRoBERTa (Barbieri et al., 2020). Transformer models have seen use in finance literature (Chen et al., 2022), and the consensus is that they have the potential, but not guarantee, of improved sentiment classification as they may capture more nuanced communication and context.[3] However, one argument against this is that simpler dictionary methods may be more robust than more complex models due to the low signal to noise ratio in text. In order to present a better overall quantification of sentiment effects, and to prevent biases from using one classifier over another, all our results consist of panels of models that incorporate sentiments from both traditional dictionaries and more advanced transformers.

Our paper builds on several recent works. Heston and Sinha (2017) use a proprietary Thomson Reuters neural network to extract positive and negative news tone for over 900,000 Reuters news articles. Interestingly, results from their study suggests that return predictability varies based on the frequency of news aggregation, as predictability from news lasts several days when aggregated daily, but from a month to a quarter when aggregated weekly. This suggests news sentiments capture information at different frequencies, and lower-frequencies may capture less transient information compared to higher-frequencies. Bybee et al. (2020) supports this, as they identify structured patterns within news text mirrors macroeconomic conditions when aggregated at the monthly frequency. In addition, their results suggest that news evolves as a somewhat separate information system, with it's own cycles of demand, shocks, and sentiment reversal patterns over

[3] See Gentzkow et al. (2019) for a summary of the mathematical motivations, and advantages and disadvantages of various methods

time. Sentiment shocks have also been investigated in economic contexts, and have been shown to have notable effects on economic consumption, production, inflation, and interest rates (Shapiro et al., 2022). Further, Binsbergen et al. (2024) shows news sentiment predicts GDP and employment growth.

Our study differs in several respects. First, we use a unique, large proprietary news dataset from Reuters consisting of over 60 million news articles over a 20 year period as the basis for our study, and by exploiting both traditional dictionary methods alongside modern transformer machine learning approaches in the classification of news text, we are able to directly compare the results from two fundamentally different approaches of deriving sentiment. Though simple dictionaries are potentially more robust, more advanced transformers have the potential for improved sentiment quantification via accounting for semantic context. Second, through a battery of different evaluation models, we explore different potential channels through which sentiment may influence future aggregate market pricing. In addition to robust linear regressions, we extend existing literature by incorporating news sentiments in several more complex models: a generalized autoregressive conditional heteroskedasticity (GARCH) model, a vector auto-regressive model (VAR), a Markov-switching VAR model, and finally a wavelet analysis.

We contribute to the existing literature as we investigate several different channels through which news sentiment effects may function. Our results are consistent with prior literature, with different sentiment classification approaches generally in agreement, and with negative news sentiment roughly twice as strong as positive sentiment(Baumeister et al., 2001). Further, we show negative sentiment effects are roughly five times stronger during periods of higher volatility Birru and Young (2022). Through GARCH modelling, we show that accounting for lagged market return variance reduces sentiment effects by roughly half, suggesting news is in-part driven by market volatility. This is further supported by VAR results, which suggests that news sentiments predict future news sentiments. Results from our two-state Markov-switching VAR analysis narrows this further, and shows that news sentiment shocks typically have short-term effects on market returns, but long-lasting effects on future news sentiments when market volatility is highest. Finally, wavelet analysis shows news clusters at frequencies of several months or lower, suggesting news sentiments capture both long and short-term investor beliefs. Overall, our study quantifies the asymmetric effects of positive and negative news sentiment shocks, highlights that news markets and financial markets are two separate information systems, and that in volatile markets, negative news

shocks result in long-lasting news pessimism, which can further destabilise markets via the social transmission channel (Hirshleifer, 2020).

The remainder of this paper is as follows. Section 2 outlines our methodology, including details of the historical news dataset, the different sentiment classification methods used, and the different models used in our evaluation procedure. Section 3 details results. Section 4 concludes.

2 Methodology

This section outlines our methodology. First, we present the details of the historical news dataset, cover the data parsing and cleaning process, and present summary news metrics. Next the different methods used to derive sentiments from the news articles are presented, along with descriptive statistics of our daily sentiment measures. Finally, we discuss details of the different models used in our evaluation procedure, along with included controls.

2.1 News Text Details

To derive measures of news sentiments over time we make use of the Thompson Reuters News Archives dataset.[4] It is an extensive global news dataset, comprised of both full news stories and news alerts in multiple languages on an intra-day frequency over the last 20 years. It contains almost 120 million news items total, with roughly 64 million from Reuters, and over 50 million from third party news suppliers. It covers both financial and non-financial news, such as news reports, company alerts, and market summaries, and spans 1st January 2000 to 31st July 2020.

Due to the extensive and complex nature of the dataset, comprehensive parsing, filtering, and cleaning was required before use. One useful advantage of the news items from Reuters is that they come pre-tagged with a series of codes from Reuters based on the contents and topics of each news item. The tagging system is extensive, with several thousand tags ranging from individual stocks, to commodities, bonds, and real-estate, to non-financial items such as politics, weather, and sport. While the tagging system does not necessarily guarantee full identification of all news items that relate to a particular topic, asset, or theme, it is extremely useful, and generally speaking fairly

[4]The financial news, data, and reporting arm of Thompson Reuters is known as Refinitiv. The News Archives dataset is sometimes referred to as Thomson Reuters MRN, machine readable news, archive.

robust. Still, the tagging of individual stocks is occasionally quite sparse for no apparent reason. For example, only a handful of individual news stories are tagged for Apple [AAPL]. Hence rather than relying on potentially sparse tagging of news stories for individual stocks, we focus on the aggregate stock market by filtering for all items pre-tagged with the S&P 500 code [SPX500]. This code is used extensively in the dataset, and captures a large amount of news coverage, ranging from simple alerts to full news summaries and articles. Opting for this relatively robust market-wide code enables us to capture a wide range of information relevant to the aggregate U.S. stock market. Note that unfortunately, third party stories do not include this tagging procedure. In addition, the content of third party items come in a variety of different formats, languages, and coverage varies sporadically over time, depending on the coverage and operation of the third party news suppliers. For these reasons, we limit our analysis to Reuters news stories only.[5]

In addition to tags, items often have multiple versions, corresponding to story updates over time, hence we filtered for the most recent version of each item chain, as prior versions could have incomplete, or incorrect information. During this process we ensured this screening was restricted to intra-day, so that any fundamental information changes from one day to another in the chain was captured, and to also ensure contemporaneous information was captured for each day over time.[6] In addition, we identified a large volume of news publications between roughly 2017 and 2019, with multiple news stories and updates roughly two orders of magnitude greater than at any other time, with updates up to the maximum number 255. While the source of these high frequency updates is not clear, they were most likely caused by changes in Reuters automated news reporting. Luckily, these items and updates were associated with several unique tags which were only present during this time in the dataset.[7] For consistency, and in an effort to eliminate noise in the dataset, we filter out items with these codes. Finally, we perform text parsing on all remaining news items' body text so it is ready for classification. This mainly involves removing headers footers, and some general formatting of article body text sections, such as removing non-relevant characters. The final cleaned dataset consists of 140,404 unique full news items, see Table 1.

Having filtered and cleaned news items tagged as relevant to the S&P 500, we present a histogram of intra-day news publication times in Figure 1. The opening hours of five exchanges are

[5]Incorporating third party news stories and alerts would take considerable time as a highly customised parsing procedure would be required to identify relevant headlines and topics. In addition, the majority of third party items come in languages other than English. For these reasons, and since more than half the news archive already comes pre-tagged from Reuters, we focus solely on Reuters news items. The potential benefit from including a relatively small number third party items is likely small to negligible.

[6]For example, the majority of version updates for a particular news item chain may have occurred on a single day, followed by an update several days later. In this case not restricting the filter to intra-day would result in timely contemporaneous news information being lost.

[7]The tags were [TOPNEWS, TOPEMRG, TOPWEALTH, TOPNOW2, TOPANZ, TOPDEALS, TOPNEWS, TOPNOW1, TOPFRX].

also shown. We observe a distinctive pattern in the data, with the majority of news articles published several hours just before, during, and after the close of New York Stock Exchange. The number of published articles greatly increases and peaks just prior to open, and spikes again just prior to markets close, after which the number greatly reduces.

Filter:	Remaining Items:	Observations Removed:
Total Number of Reuters News Items	64,587,440	
Remove non-English items	42,425,573	22,161,867
Items that include S&P 500 tag	475,748	41,949,825
Remove unavailable, cancelled, deleted items	475,715	33
Remove noise from dataset*	198,557	277,158
Select latest version for chained items*	153,568	44,989
Total Alerts (headline text only)	13,164	
Total Articles (headline & body text)	140,404	

Table 1: Reuters news filtering summary.
Summary of filters applied to the Reuters News Dataset. Data coverage is from 1st January 2000 to 31st July 2020. *Note we remove extensive noise patterns that appear in the dataset between roughly April 2017 to 2019. This noise consisted of a large number of highly frequent news reports and updates, and were most likely due to a change in Reuters automatic news reporting methodology, which was subsequently discontinued.

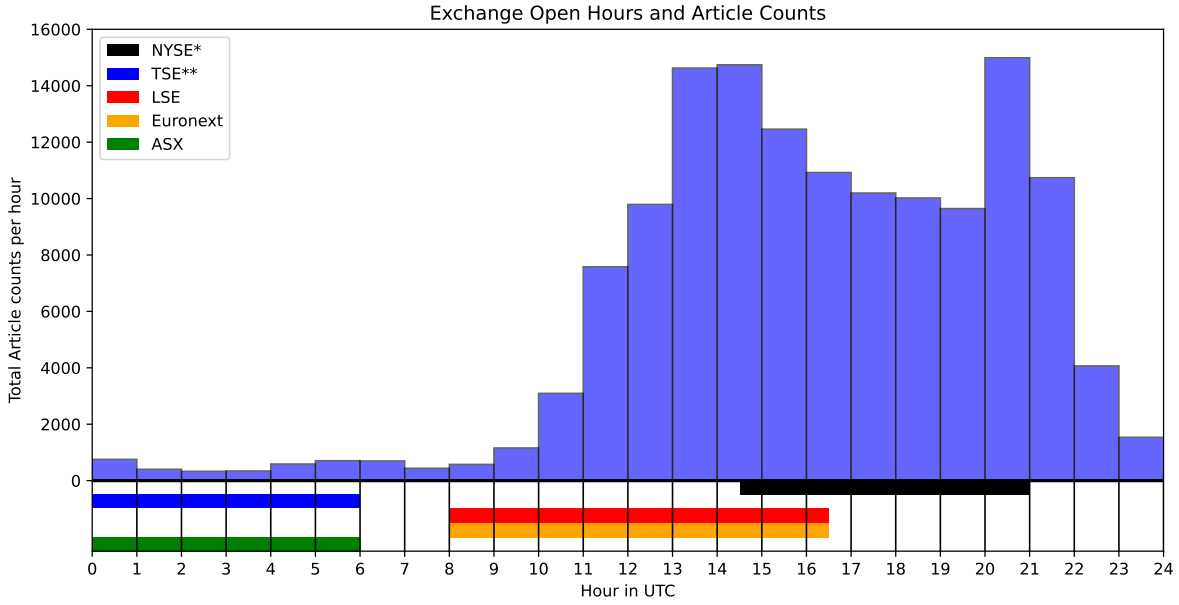


Figure 1: Histogram of filtered S&P 500 news publications per hour in UTC. Counts represent total counts over the 20 year study period. Default operating hours of major exchanges are also shown, as indicated by individual exchanges at the time of writing. Note actual open hours may vary slightly over time, e.g. depending on local daylight savings times. *New York Stock Exchange, NYSE, also includes NASDAQ open hours. **Tokyo Stock Exchange, TSE, has lunch from UTC 02:30 to 03:30.

To further illustrate news story volumes over time, Figure 2 shows daily S&P 500 news story counts, along with daily changes in the number of news updates, the daily % change in S&P 500 returns, and the daily VIX index at market close. Recession periods based on the NBER recession indicator [USREC] from the Federal Reserve Bank of St. Louis (2022) are also plotted.

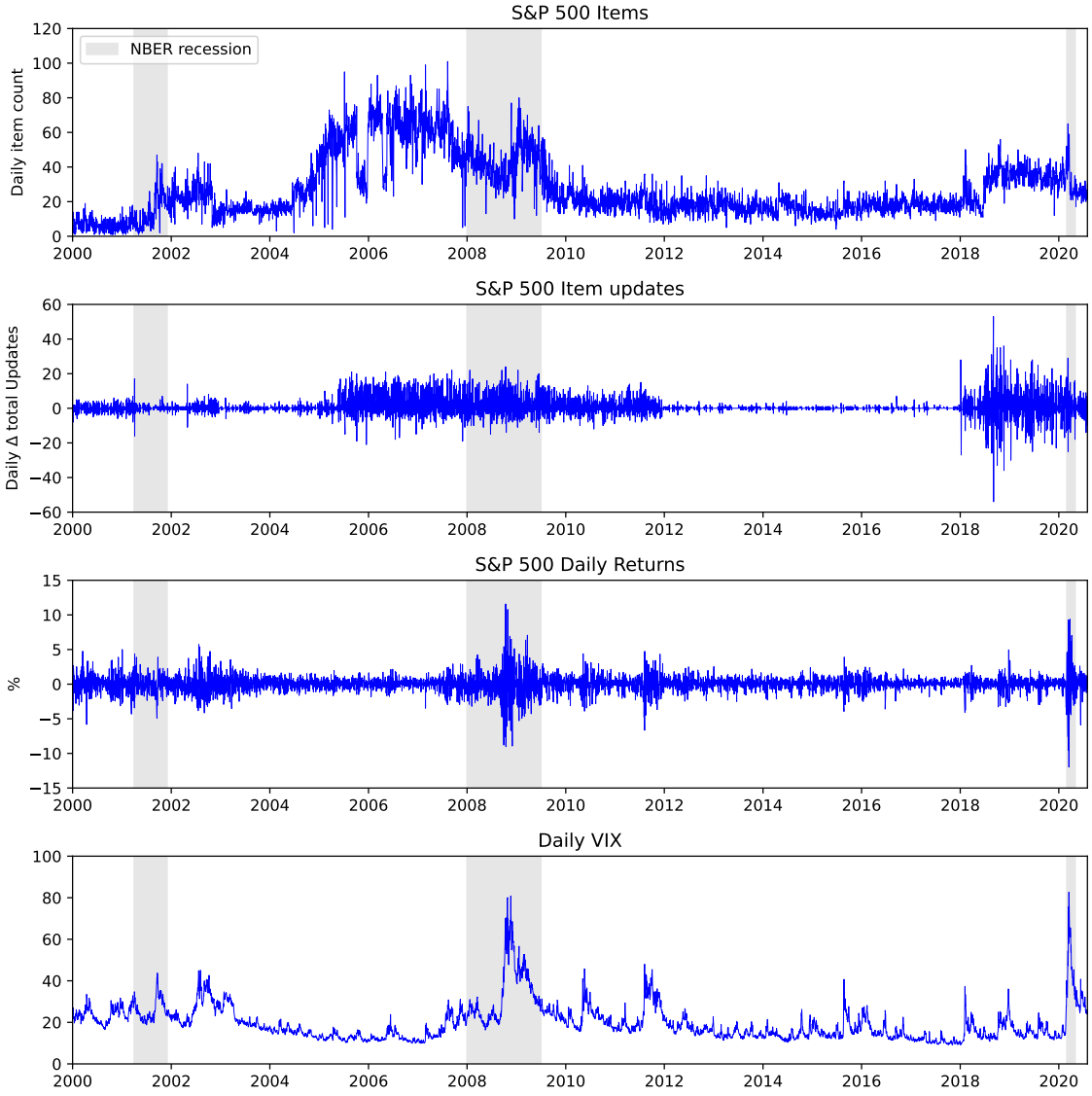


Figure 2: The first plot (top) shows the daily total number of filtered news articles tagged S&P 500 from 1st January 2000 to 31st July 2020. The second plot shows the daily change in the number of news story updates per day. The third plot shows the daily returns of the S&P 500. The fourth plot shows the daily VIX at market close.

Several patterns emerge. Figure 2 plots 1 and 2 show that the total number of news items and updates remain relatively consistent over time. However, during the lead up to the 2007-09 Global Financial Crisis (GFC), the number of stories rises, and remains elevated during this time,

only returning to pre-2004 levels after 2010. Sharp spikes are also observed at other points in time, including just after the September 11 2001 terrorist attacks and the collapse of Enron which occurred the following month, after which subsequent news volumes remained elevated for roughly a year. Another spike is observed in March of 2020, coinciding with when western news outlets started to publish concerns about potential fallout from the COVID-19 virus coming out of China.[8] A small spike in news publications is also observed in 2002, possibly in-line with the WorldCom scandal, along with an increase in news updates in roughly 2018, the cause of which is unclear, but is most likely attributed to changes in Reuters news methodology. Similar patterns are observed with the changes in the volume of daily news updates, with increased updates during the years of the GFC, and an increase post 2018.

Overall we observe news spikes and elevated levels of news coverage and updates align with periods of recession, or around major financial or world events. Our results are consistent with prior literature, and illustrate that news follows cycles, with news media giving more attention to specific topics and themes during certain times (Bybee et al., 2020; Mullainathan and Shleifer, 2005).

2.2 Sentiment Classification

Having filtered and cleaned the news articles, we next classify the sentiment of each news article individually. As discussed, for robustness we use five different sentiment classifiers to derive both positive and negative sentiment tones for each article separately. We use three traditional dictionary approaches, and two more advanced transformer neural-networks.Once all 140,404 individual news items are classified, we then calculate the daily average of positive and negative sentiments for each classifier separately, and take these as our daily sentiment measures. Each sentiment classifier is detailed below.[9]

[8] Interestingly, mainstream western news was generally slower to react than non-mainstream news sources, even though negative news and mass deaths were identified in China at least one months earlier.

[9] Note we did not use large language models such as GPT due to vendor reservations about potential data retention, etc. For the two transformer neural networks, two locally running Nvidia GPUs were used to speed up the process.

Loughran and McDonald

The first classifier is the Loughran and McDonald (2011) financial keyword dictionary. Based on their dictionary, the proportion of positive and negative keywords present in an article are calculated. Daily measures of positive and negative sentiments are then calculated via averaging these daily proportions separately. We also include a negation check on positive words to prevent false positives, based on if one of the three prior keywords is a negation word, for example "not good".[10] Given positive, negative, and negation wordlists P, N and X respectively, for n articles on a particular day, each with m total words, w, the daily positive LM_P and negative LM_N sentiment measures are defined as,

$$\begin{aligned} LM_P &= \frac{1}{n}\sum_{j=1}^{n}\frac{\sum_{i=1}^{m}\mathbb{1}\left\{\begin{array}{l} w_i \in P \\ w_{i-1},...,w_{i-3} \notin X \end{array}\right\}}{m} \\ LM_N &= \frac{1}{n}\sum_{j=1}^{n}\frac{\sum_{i=1}^{m}\mathbb{1}\{w_i \in N\} + \sum_{i=1}^{m}\mathbb{1}\left\{\begin{array}{l} w_i \in P \\ w_{i-1},...,w_{i-3} \in X \end{array}\right\}}{m}. \end{aligned} \tag{1}$$

VADER

The second classifier is the VADER (VA) sentiment analyser by Hutto and Gilbert (2014). Part of Python's Natural Language Toolkit, VADER is a human-curated lexicon comprised of a weighted dictionary, that loosely takes into account word intensity and context. The default *vader_lexicon* was used as it is widely used and shown as applicable across several different domains. VADER consists of outputs probabilities of the story being classed as positive, neutral, negative, and in addition gives an overall polarity score of most positive to most negative. We apply VADER to each article separately, and take the daily averages of the positive and negative probability scores as our sentiment measures. For VADER function positive and negative output probabilities V_P and V_N, for n articles on a particular day, the sentiment measures are calculated as,

$$VA_P = \frac{1}{n}\sum_{i=1}^{n} V_{i,P} \qquad VA_N = \frac{1}{n}\sum_{i=1}^{n} V_{i,N}. \tag{2}$$

[10] See Appendix B.1 for a list of negation keywords. Note we do not a negation check on positive words, as these are less common, (Loughran and McDonald, 2011).

FinBERT

The third classifier FinBERT (FB) (Araci, 2019) is a modified version of the BERT transformer neural network. Trained on a human labelled financial phrasebank, it has been fine-tuned to classify sentiment of financial-specific text. By default FinBERT outputs a classification label; positive, neutral, or negative, along with associated probability, or softmax score. Rather than just using this single score, we extract the raw positive and negative logits for all articles, and use the daily average as our daily sentiment measures, similar to Equation 2. This is done as we wish to extract both positive and negative sentiments from each news item. In addition rather than using the softmax probabilities, the raw logits are more granular and have infinite support. Also, since FinBERT has three classes, we avoid collinearity issues.[11]

One aspect worth noting is that FinBERT input text is limited to 512 tokens, hence we perform manual truncation if text exceeds this limit. While the majority of news stories are longer than this limit, we investigate several variations of inputs, including 50-50 first-last half of tokens as input, and batching all text and averaging across each individual news article. However the results showed little difference in summary metrics. Also, generally speaking the most important information in news stories is listed first, and since the typical approach in such cases is to use the first tokens and truncate the rest, we adopt this approach.

TwitterRoBERTa

The fourth classifier is the TwitterRoBERTa (TR) transformer neural network from (Barbieri et al., 2020). It is a RoBERTa based BERT model trained on approximately 58 million social media Twitter tweets, and fine-tuned for sentiment analysis with the TweetEval benchmark. As with FinBERT, output consisted of softmax classifications of positive, neutral and negative. For the purpose of aggregation, the same raw positive and negative logits were averaged daily as our sentiment measures. TwitterRoBERTa also has the same limit around input text, and we use the same manual truncation method in the parsing process.

[11] Raw logits are taken from the second last network layer before softmax is applied. While there is a second FinBERT model (Yang et al., 2020), at the time of writing there are concerns with this model, as the probability classes sum to greater to one. For this reason we use the Araci (2019) model.

KE Top 100

The fifth and final classifier is based on a dictionary approach using the top 100 keywords identified by Ke et al. (2019), (KE). This final method was chosen as an exploratory exercise. While the LM dictionary of keywords is widely recognised, it may be that specific words are particularly relevant and dominate sentiment classification. Hence we use this shortlist of top 50 positive and 50 negative words to classify text sentiment. The approach is the same as Equation 1, but without the negation check as it is not applicable for some keywords. Proportions of positive and negative words are calculated for each news story separately, and then averaged for each individual day.[12]

2.3 Daily Sentiments Summary

Summary statistics of aggregate daily sentiments from each classifier, and contemporaneous S&P 500 daily % returns are presented in Table 2. Reassuringly, all positive (negative) measures are positively correlated with other positive (negative) measures, and inversely correlated with their counterparts. The strongest correlations occur between the two transformer models, while some of the weaker tone correlations are associated with KE, likely due to the limited dictionary used. Results between positive and negative measures and contemporary market returns are in agreement with what one would expect.

		M	SD	1	2	3	4	5	6	7	8	9	10
1. LM	(+)	0.0112	0.0047	-									
2. VA	(+)	0.0855	0.0159	0.5	-								
3. FB	(+)	-0.3781	0.8270	0.63	0.52	-							
4. TR	(+)	-0.2865	0.7963	0.74	0.57	0.9	-						
5. Ke	(+)	0.0022	0.0014	0.39	0.25	0.32	0.35	-					
6. LM	(-)	0.0228	0.0075	-0.32	-0.47	-0.58	-0.64	-0.17	-				
7. VA	(-)	0.0571	0.0199	-0.32	-0.41	-0.52	-0.62	-0.13	0.79	-			
8. FB	(-)	1.3643	1.0181	-0.55	-0.5	-0.97	-0.86	-0.27	0.61	0.56	-		
9. TR	(-)	-0.6758	0.8385	-0.71	-0.56	-0.91	-0.99	-0.34	0.68	0.65	0.88	-	
10. Ke	(-)	0.0065	0.0035	-0.37	-0.39	-0.69	-0.62	-0.19	0.56	0.49	0.72	0.65	-
11. S&P 500 Ret	(%)	0.0240	1.2560	0.36	0.31	0.57	0.53	0.18	-0.34	-0.29	-0.55	-0.54	-0.4

Table 2: Summary statistics of daily sentiments and contemporaneous S&P 500 daily % returns.
This table presents means, standard deviations, and Pearson correlation coefficients for daily positive and negative sentiments from each classifier, and contemporaneous daily % returns of the S&P 500. $n = 5,161$.

Next, we illustrate distributions of daily positive and negative news sentiments in Figure 3. For clarity, the three traditional dictionary approaches LM, VA, KE are grouped together, along

[12]Negation on positive words is not used as it does not make sense for several words, such as treasury, or reconnaissance. Since many of the 100 words identified are not part of the LM dictionary, it is interesting to compare the classifications and predictive accuracy of the KE to LM.

with transformers FB and TR separately. The resulting histograms are generally similar for both positive and negative measures. For positive news, all histograms are slightly positively skewed, however for negative news the transformers are negatively skewed, the opposite of LM, VA and KE. This opposite skewness of the transformer methods is most likely due to the fact they classify each story as positive, neutral, or negative. Since these probabilities sum to one, and since the transformers typically assign a very high probability to one of the classes for each story, the result is that both positive and negative sentiments are skewed in opposite directions. By comparison, positive and negative word counts of dictionary methods are independent, and hence do not have such extreme skew.

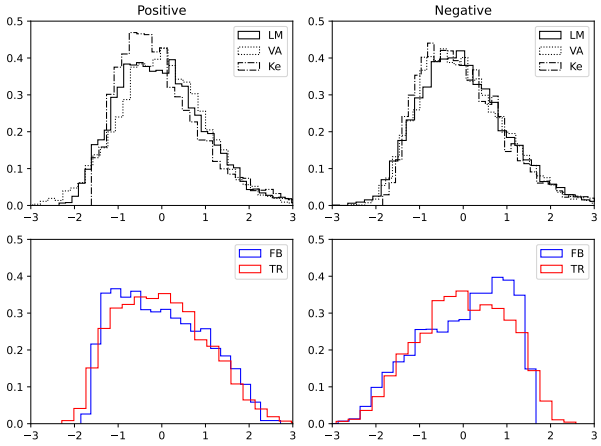


Figure 3: Histograms of standardized daily positive and negative sentiment measures for each classifier.

Further, Figure 4 shows scatter plots of standardized daily aggregated positive and negative sentiment pairs for each classifier. One striking difference between dictionary and transformer models is that the bivariate point clouds of the dictionaries are much more dispersed. By comparison, the transformer model point clouds are mainly distributed diagonally, indicating that, when aggregated, these positive and negative measures are mostly inversely correlated. Though this may be somewhat expected, as days with more negative news may have less positive news, the differences in point clouds of these two different classifier types is striking. The results indicate the two different classifier classes are interpreting the text information differently, with transformers likely less capable of capturing a wider range of sentiments and hence investor beliefs. This lessened flexibility in sentiment classification could hinder model performance due to potential collinearity, and could hinder the identification of the relationship between news sentiment and market returns.

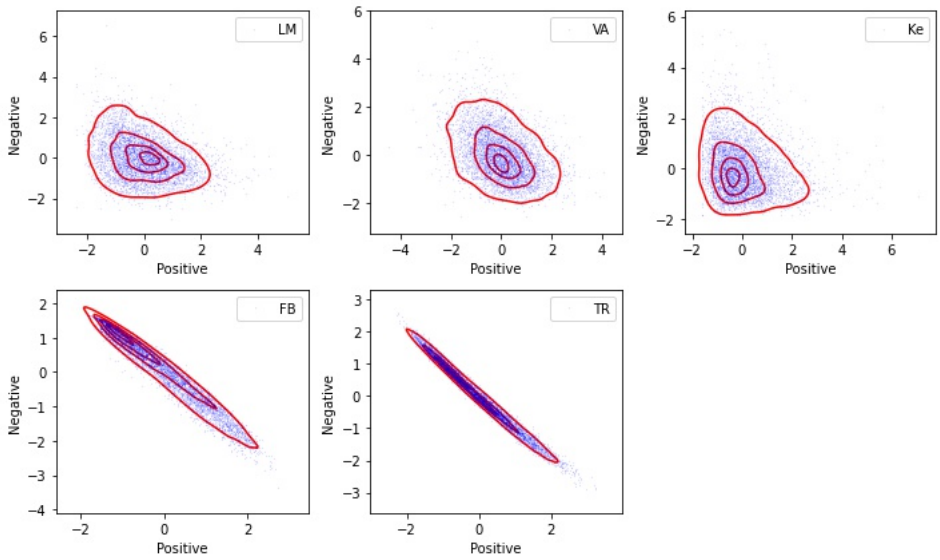


Figure 4: Scatterplots of standardized daily positive and negative sentiment pairs for each classifier, along with contour plots at 10, 50, 85, and 95% intervals.

Finally, we compare news sentiments over time to investor sentiments as proxied from the American Association of Individual Investors (2022), (AAII). Each week, AAII surveys a group of registered individual investors, and aggregates these reports into consensus on the belief of the future direction of the aggregate stock market over the next six months as either bullish, neutral, or bearish. Six month rolling averages of the bullish and bearish measures from AAII, and positive and negative news sentiments are presented in Figure 5.

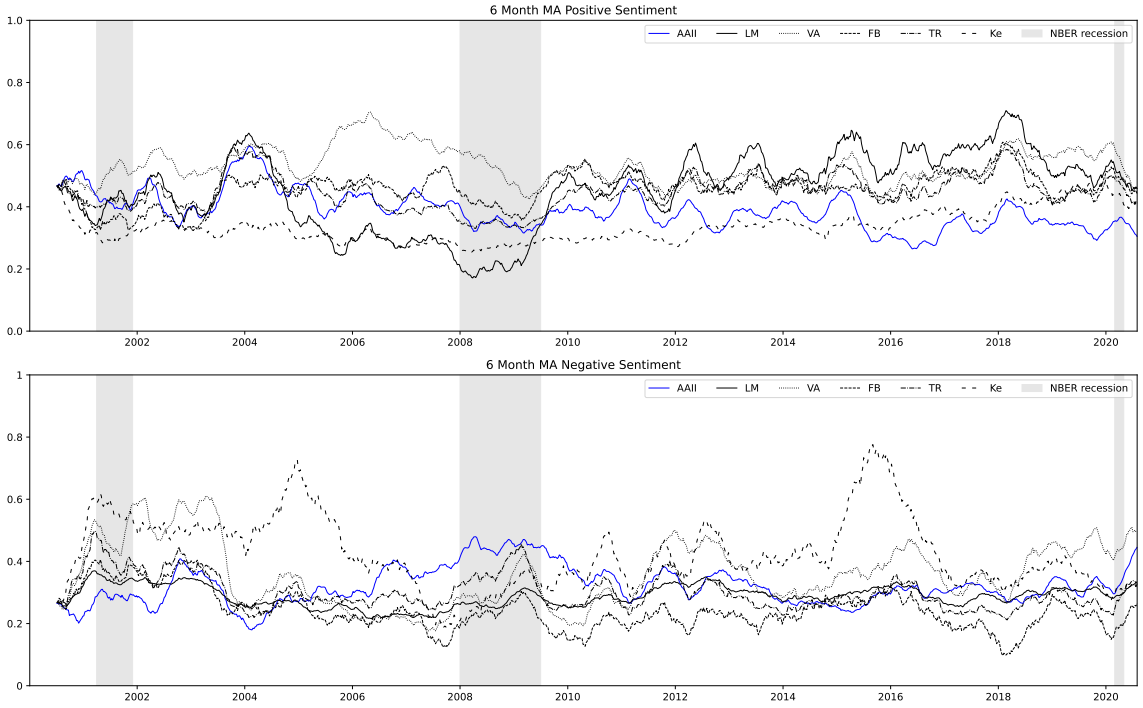


Figure 5: Six month moving averages of positive and negative news sentiments and bullish and bearish business conditions from AAII. For easier visualisation, all series have been vertically shifted to start at the same initial value as the AAII series, and due to the low proportions of positive and negative words of dictionary methods, the moving averages of LM ,VA, and KE are vertically scaled by a factor of 50, 10 100, for positive, and 10, 10 100 for negative sentiments respectively.

We observe generally the series have similar peaks and troughs over time, with most series tracking closely with each other as peaks and troughs often closely align. For example, we observe a peak of positive news in 2004, and several peaks and troughs through 2011 to 2016. While negative news from text does not align with AAII as closely as positive news, we do observe some similarity, for example with increased negative news in the 2008 GFC period, and with the onset of COVID-19 in early 2020. Interestingly, LM tracks positive AAII sentiment most closely, however is relatively flat for negative news, suggesting that while LM may be robust in identifying positive news, it may not necessarily be as robust at capturing negative news. For both FB and TR transformer classifiers, positive and negative news sentiments somewhat track the AAII data. We observe a drop in positive news during the last quarter of 2001, and a reduction (increase) in positive (negative) news in the 2008-2010 period. Also, several local peaks and troughs in AAII appear to be followed closely by sentiment from transformers. Differences between AAII sentiment and news sentiments may be due to several reasons, such as the methodology AAII use for data capture, or the fact they capture data at lower weekly frequency compared to our daily measures. Still, the plots illustrate that generally our news-based sentiments are not dissimilar to investor sentiments proxied via surveys.

2.4 Evaluation Procedure

To determine the predictability of market returns from news sentiments, we evaluate several different types of regression models, Each regression models next day % returns of the S&P 500 based on lagged positive and negative sentiments, and a series of controls. Our analysis is primarily on the daily frequency as short-term sentiment effects, such as momentum and potential overreaction, are likely most detectable at this frequency.[13] Each model consists of a panel regression, were the sentiments from each classifier are included as separate regressions as separate columns. For ease of comparison, each panel regression also includes a control regression comprised of controls only.

Each panel and regression model is comprised of up to two lagged periods, or days, of positive and negative sentiments, and a series of controls. We limit the number of lags to two because short-term effects and mispricing is most likely to occur within a few days of news release. In addition, while we could have included more lags, this risks overfitting and suppression effects via the inclusion of additional variables. Preliminary analysis suggested additional lagged terms were not significant, and given the trade-off of overfitting and suppressor effects, we select two days of lags for our approach. Contemporaneous news, controls, and sentiment measures were deliberately excluded from all models, as contemporaneous innovations from news and controls are likely to have an instantaneous effect or co-movement with the daily stock market price. Hence only lagged controls are incorporated into our models.

We control for past daily: S&P 500 % returns, the VIX index at market close, and we include three measures of short-term, medium-term, and long-term cost of credit, or credit spreads. S&P 500 daily close price data and CBOE Volatility VIX index data is taken from Refinitiv (2021). Credit spread data is from the Federal Reserve Bank of St. Louis (2022). TEDRATE is the short-term spread, calculated as the difference between 3 month LIBOR in USD [USD3MTD156N], and 3-Month Treasury Bill rate [DTB3]. TERMSPREAD is the medium term spread calculated from the daily spread between 10-Year Treasury Bond rates [T10YFF], and 1-Year Treasury Bond rates [T1YFF]. BAA10Y is the long term spread between Moody's Seasoned BAA Corporate Bonds of 20 years plus, and 10 year Treasury Constant Maturity [BAA10Y]. Non-trade dates and holidays were not padded, but rather variables lagged relative to each trading day.[14]

[13] Though our main focus is on the daily frequency, sentiments may have longer lasting effects at lower frequencies, and we explore this via a wavelet analysis, see Section 2.4.

[14] We also conducted preliminary investigations using a weekend dummy, but as this did not drastically change findings, was excluded.

One conceivable complication of using regression models with both positive and negative measures of sentiment tone is potential collinearity. If positive and negative sentiments are perfect linear combinations of each other, any regression using both would result in the coefficient of one reducing to zero. For example, in the case of a binary classifier with only two classes. This is the main reason we do not use BERT, or other binary classifiers. The five classifiers we use do not suffer from this issue, as the proportional positive and negative word counts of LM and KE are independent, and VA, FB, and TR all have three classes; positive, neutral, and negative. However, it is worth noting Table 2 which indicates daily positive and negative sentiment measures of both FB and TR are strongly inversely correlated, so we might expect one of the sentiment variables to dominate the other in regression results.

OLS Model

Our first analysis is based on a simple OLS regression on the daily % returns of the S&P 500. Separate regressions are run using lagged sentiment pairs from each classifier. Two regressions are run for each classifier, with one and two periods of lagged sentiments used respectively. The model is defined as,

$$R_t = \alpha + \sum_{t=-1}^{-2} \mathbf{V_t} + \varepsilon_t \qquad \text{where} \qquad \mathbf{V_t} = \begin{bmatrix} S_p \\ S_n \\ R \\ Tedrate \\ Termspread \\ BAA10Y \end{bmatrix}, \tag{3}$$

where $\mathbf{V_t}$ is a vector of lagged daily positive sentiment S_p, negative sentiments, S_n, % changes in the S&P 500 index R, and lagged controls Tedrate, Termspread, BAA10Y. Residuals are assumed normally distributed.

Markov-switching OLS Model

Prior literature suggests sentiment effects may differ based on the state of financial markets, and that sentiment effects may be dependent on market volatility, particularly during times of crisis and greater market volatility (Birru and Young, 2022). Hence we use a Markov-switching OLS regression model that follows Equation 3, but allows for multiple states, or regimes, within the dataset. That is, the model internally allocates each observation to a particular state, based on model specification and the number of states given by the user. We specify two states, corresponding

to a low volatility state, and a high volatility state. This enables us to identify potential differences in sentiment significance and magnitude during times of low (high) volatility separately.[15]

The model simultaneously minimises the mean-square error of linear regressions across both states, while maximum likelihood estimation defines the probabilities each data point is in either the low, or high volatility state. All sentiments and controls are considered exogenous and included in both states. Each regime is similar to Equation 3, but also includes an extra constant variance term σ^2 for each state. Intercept and variance terms are both considered switching, with no targeting used. The models associated transition matrix is given by Equation 4, where transition matrix values 0 (1) represent low (high) variance regimes,

$$\mathbb{P}(S_t = s_t | S_{t-1} = s_{t-1}) = \left[\begin{smallmatrix} p_{00} & p_{01} \\ p_{10} & p_{11} \end{smallmatrix}\right]. \tag{4}$$

GARCH Model

To further disentangle and evaluate potential effects of sentiment from volatility, we follow in the spirit of Engle and Bollerslev (1986), and use a generalized autoregressive conditional heteroskedasticity (GARCH) model. GARCH models are useful in describing systems where heteroskedasticity, or unequal variance over time, is present. The standard GARCH model performs least squares minimisation in the same manner as Equation 3, however in addition the model conditions the variance as dependent on prior variance and error terms.[16] Compared to a Markov-switching model which allows modelling over different regimes, GARCH allows us to account for return conditional variance, and hence identify if heteroskedasticity subsumes sentiment effects, and the persistence of volatility.

The model follows the same mean equation as OLS model Equation 3, however rather than assuming normally distributed error terms and variance, the conditional variance term of standard GARCH is modelled as follows,

$$\sigma_t^2 = \omega + \sum_{j=1}^{q} \alpha_j \varepsilon_{t-j}^2 + \sum_{j=1}^{p} \beta_j \sigma_{t-j}^2, \tag{5}$$

[15] A two-state model is simplest and most defensible. Given markets are often described as rational or irrational, and since volatility clustering is a well-known phenomenon, two states makes the most sense. More states also increase the chances of overfitting and convergence issues, and while we initially investigated a three-state model alternative, these models consistently failed to converge.

[16] While there are many families of GARCH models, such as exponential eGARCH, and threshold tGARCH etc, we use the standard GARCH model for it's simplicity, and robustness compared to more exotic models. Also, the theoretical underpinning of more exotic models are harder to justify in practice.

where ω is the variance intercept constant, p the order of lagged variance terms, and q the order of lagged residuals. We follow simple GARCH, and specify GARCH order $p, q = 1, 1$ as this is one of the most common specifications, and is simplest to interpret and defend.[17] Higher order GARCH models can have instability, and are harder to explain in practice. Hence in practical GARCH applications, variance and residual orders remain low. Volatility persistence is determined by the magnitude and sum of lagged variance equation coefficients α_i, β_i, and they must sum to less than one for variance to decay, otherwise the system is non-stationary.

VAR Model

Next we use a vector autoregression (VAR) analysis to investigate potential relationships between returns and sentiments. The model solves a multivariate time series regression, where each variable is modelled as a $t-i$ linear regression of other variables and controls, with the system of equations solved simultaneously. VAR allows us to capture potential autoregressive effects for both positive and negative news sentiments and returns separately, while also capturing how these variables affect each other. Given a multivariate time series vector $\mathbf{Y}$ and K variables, or in other words K equations where each equation is a variable dependent on all others, the VAR model solves all K simultaneous equations of T observations each. For consistency, our VAR model includes market returns, sentiments, and the same controls as Equation 3, and we specify lag order 2. Hence the VAR model is given by,

$$\mathbf{Y_t} = \mathbf{A_1 Y_{t-1}} + \mathbf{A_2 Y_{t-2}} + u_t, \qquad \text{with} \qquad u_t \sim N(0, \sigma_u), \tag{6}$$

where $\mathbf{A_i}$ is a $K \times K$ coefficient matrix.

Markov-switching VAR Model

One limitation of the standard VAR model is that it does not take into account that the relationships between news sentiments and returns may vary at different times, such as during periods of greater market volatility. For example, Figure 2 shows volatility clustering, and increased news occurs around the time of the GFC. In contrast, while the Markov-switching OLS regression model takes

[17] As a side note our mean equation is equivalent to ARMRA, auto-regressive moving average, order $p, q = 2, 0$, or in other words, two days of auto-regressive lagged returns, with no moving-average terms.

into account that sentiment effects may vary during times of higher volatility, it does not capture potential autocorrelation effects of sentiments, nor the potential effects that market returns may have on news sentiments. Hence we turn to a Markov-switching VAR model to aid in our analysis.

The model is effectively a combination of Equations 4 and 6, and has the potential to capture both differences in short and long-lasting effects between returns and both positive and negative sentiments, and vice versa, during times of low and high market volatility. The model uses a Sims-Zha prior, Gibbs sampling, and block optimisation in the following order, maximise over: the intercepts, the $AR(p)$ coefficients, the error covariances Σ, the transition matrix Q.

We include all controls in the model, and specify one period of lag to aid in model convergence. We specify two regimes based on changes in S&P 500 returns, to ensure results correspond to a low volatility state, and a high volatility state.[18] Though the model does not output equation coefficients for each regime, it outputs regime switching rates, smoothed low (high) regime probabilities of each day, and impulse response functions (IRF) generated from model simulated low (high) states. These IRF's represent the estimated response in one variable given a one-unit change in another, and by comparing IRF's for returns and sentiments across both low and high volatility states, we are able to quantify the effects of sentiment on returns and vice versa, when markets are both less and more volatile.

Wavelet Model

Finally, we use a wavelet analysis to investigate if long-running contagion effects exist between news market sentiments and the U.S. equity market. Since news markets are in part driven by criteria other than pure financial fundamentals, such as reader demand and belief reinforcement (Mullainathan and Shleifer, 2005), news sentiments may not always correlate with fundamental financial information. Hence the flow of news information and coverage of particular topics and themes may capture information such as increased pessimism beyond the scope of traditional financial market information and fundamentals (Bybee et al., 2020; Bybee et al., 2024). Particular topics and themes may vary over time, and long-running pessimism may last for some time after large financial shocks. Hence extended periods of negative news may prolong market downturns.

[18] Further hyperparameters include; lags=1, regimes=2, prior tightness=0.3 (0 loosest), tightness of prior around AR(1) parameters (0 loosest), lag decay=1, tightness around intercept=0.5, tightness around exogenous variable coefficients=0.1, lag decay equivalence set to monthly, and 20,000 iterations for each model.

We perform a wavelet coherence analysis, which has the potential of identifying long-lasting contagion effects between news market sentiments and U.S. equity market returns. Figure 2 suggests market recessions and heightened news coverage coincide, hence contagion effects are most likely to arise during times of financial crisis (Allen and Gale, 2000; Akhtaruzzaman et al., 2021). Wavelets have been used to identify cross-market contagion (Dewandaru et al., 2016; Gallegati, 2012), and hence we adopt a wavelet analysis as it allows us to quantify news sentiment and U.S. equity return correlations across both time and multiple frequencies simultaneously.

One of the primary metrics is the squared cross-wavelet coherence $R^2(n,s)$, or wavelet coherence for short. Bound $[0,1]$, it is a measure of the local correlation coefficient between two series $x(n)$ and $y(n)$ in time-frequency domain. It is calculated as,

$$R^2(n,s) = \frac{|S\big(S^{-1}W^{XY}(n,s)\big)|^2}{S\big(S^{-1}|W^X(n,s)|^2 \cdot S^{-1}|W^Y(n,s)|^2\big)} \quad \text{with} \quad W^{XY}(n,s) = W^X(n,s)\overline{W^Y(n,s)} \tag{7}$$

where S is a time-scale smoothing operator, n the time index, and s the time scale, or period, which is inversely proportional to the frequency. We perform wavelet coherence analysis on market returns and positive and negative sentiments separately, to produce heatmaps showing $R^2(n,s)$ values over time for different frequencies.

3 Results

3.1 OLS

Results from OLS regressions Equation 3 are show in Table 3. Lagged sentiments are significant across most models, with generally only one of either positive or negative sentiment significant. This is likely due to collinearity, which results in one sentiment tone dominating the other. Overall negative sentiment almost always has both a larger coefficient magnitude, and is more significant than positive sentiment, in agreement with existing literature of bad news being stronger than good (Baumeister et al., 2001).

	Control	LM 1	LM 2	VA 1	VA 2	FB 1	FB 2	TR 1	TR2	KE 1	KE 2
P_{t-1}	-	0.0618	0.043	0.0007	0.0008	0.0023 ***	0.0023 ***	-0.0004	-0.0008	-0.0779	-0.0922
N_{t-1}	-	-0.0866 ***	-0.0978 ***	-0.0263 **	-0.03 ***	0.0011 *	0.0011 *	-0.0014	-0.0017	-0.188 ***	-0.1863 ***
P_{t-2}	-	-	0.0512	-	-0.0017	-	0.0005	-	0.0013	-	0.0727
N_{t-2}	-	-	0.0312	-	0.009	-	0.0003	-	0.0013	-	-0.0102
R_{t-1}	-0.0692	-0.0934 **	-0.0938 **	-0.0809 *	-0.0819 *	-0.1095 **	-0.1096 **	-0.1055 **	-0.1055 **	-0.0894 **	-0.0895 **
R_{t-2}	-0.0202	-0.0313	-0.0312	-0.0261	-0.0232	-0.0275	-0.0292	-0.0306	-0.0292	-0.0239	-0.0259
VIX_{t-1}	0.0429	0.0441	0.0435	0.0435	0.0432	0.0385	0.0389	0.0399	0.0398	0.0411	0.0409
VIX_{t-2}	-0.0368	-0.0357	-0.0352	-0.0354	-0.0357	-0.0317	-0.032	-0.0318	-0.0321	-0.0352	-0.035
$Tedrate_{t-1}$	0.0078	0.008	0.0083	0.0077	0.0078	0.0081	0.0081	0.008	0.0081	0.0078	0.0078
$Tedrate_{t-2}$	-0.0094	-0.0098	-0.0099	-0.0097	-0.0097	-0.0095	-0.0094	-0.0096	-0.0097	-0.0096	-0.0096
$Termspread_{t-1}$	0.0061	0.005	0.0049	0.0057	0.0056	0.0046	0.0046	0.0047	0.0047	0.0051	0.0052
$Termspread_{t-2}$	-0.0063	-0.0053	-0.0052	-0.006	-0.0058	-0.0049	-0.0049	-0.005	-0.005	-0.0053	-0.0054
$BAA10Y_{t-1}$	0.0054	0.0062	0.006	0.0061	0.006	0.0055	0.0055	0.0059	0.0059	0.0055	0.0056
$BAA10Y_{t-2}$	-0.0054	-0.0061	-0.0059	-0.006	-0.0059	-0.0054	-0.0054	-0.0058	-0.0059	-0.0055	-0.0055
Intercept	0	0.0007	0	0.0011	0.001	-0.0009	-0.0011	-0.0014	-0.0004	0.0013	0.0012
R Squared	0.0194	0.0226	0.023	0.021	0.0212	0.0232	0.0233	0.0222	0.0224	0.0219	0.022
Adj. R squared	0.0175	0.0203	0.0204	0.0187	0.0185	0.0209	0.0206	0.0199	0.0197	0.0197	0.0194
AIC	30,601	30,614	30,612	30,606	30,602	30,617	30,614	30,612	30,609	30,611	30,607
BIC	30,522	30,522	30,508	30,514	30,498	30,525	30,509	30,520	30,504	30,519	30,502
n						5,161					

Table 3: OLS regression results.
Regression results from Equation 3, which fits next day S&P 500 % returns from lagged news sentiment measures and controls. Two regressions are performed for each classifier, with one and two days of lagged sentiments respectively. A regression consisting of controls only is included for comparison. Robust standard errors used. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Negative sentiment effects are roughly twice as large as positive sentiment effects, and coefficient signs suggest an increase in negative news from the previous day results in decreased next-day returns, and vice versa for positive news. Interestingly, these signs reverse for news sentiments from two days ago, reflecting momentum reversal patterns observed in prior literature (Birru and Young, 2022). Coefficients of lagged prior returns are also negative, suggesting return reversals, with coefficients largest and significant for the prior day only.

While all models have improved adjusted R-squared and AIC values compared to the control regression, only a few models have improved, i.e. larger BIC values. Generally adjusted R-squared values are greatest for models with only one period of lagged sentiment, with LM and FB models performing the best, likely reflecting the robustness of the LM dictionary, and the ability of the FB neural network to take into account semantic context. Combined with coefficient significance of the various models, the results give credence to our choice to restrict models to one or two days of lagged sentiments.

Interestingly, FB models show high significance for positive news on the previous day, while negative news dominates across all other models. This points to FB identifying particularly important positive information. In comparison, TR models perform the worst with no sentiment coefficients significant, though the negative news coefficients are roughly double that for positive sentiments. This is likely as TR was not trained for finance-specific domain classification. Also interestingly, KE outperforms VA, which highlights that some of the shortlist keywords are particularly strong, and capture relatively more information.

3.2 Markov-switching OLS

Results from the two state Markov-switching OLS model (MS-OLS) Equation 4 are detailed in Table 4. Model smoothed probabilities of days being classed as in a low or high volatility state over time are shown in Figures 6 and Figure 7, which show the full sample period, and last year of sample respectively.

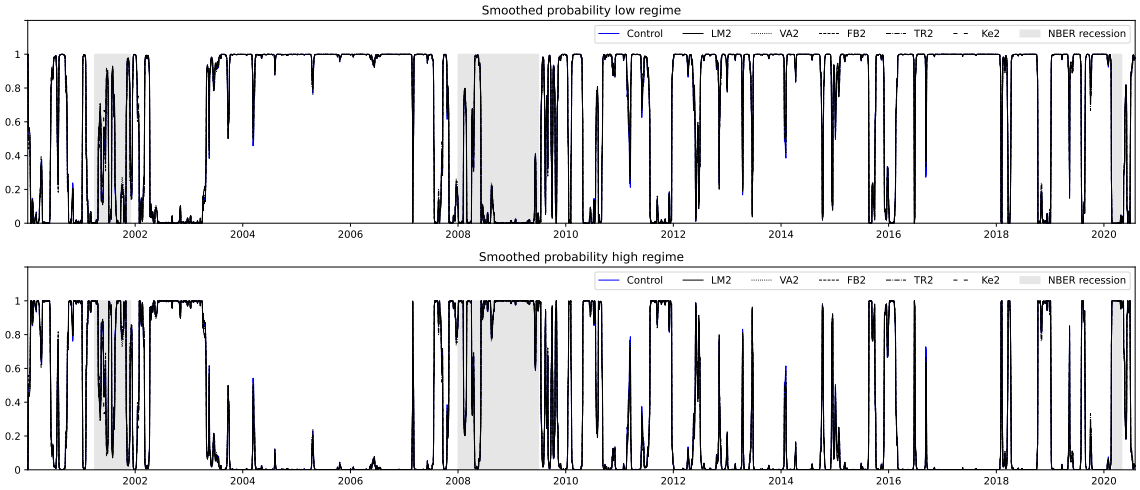


Figure 6: Plots of smoothed probabilities of the S&P 500 being in a low (top) and high (bottom) volatility state over time, for the full sample period 1st January 2000 to 31st July 2020. While switching occurs at various times throughout the sample, notable periods of sustained high volatility align closely with periods of recession per the NBER recession indicator, such as late 2001, around the 2008-2009 GFC, and at the start of COVID-19 in 2020.

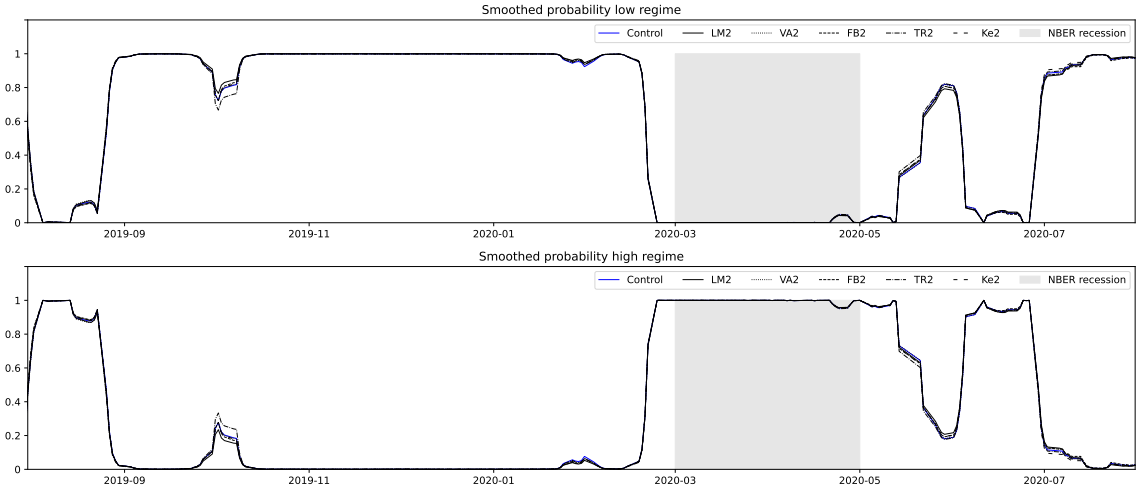


Figure 7: Plots of smoothed probabilities of the S&P 500 being in a low (top) and high (bottom) volatility state over time, for the last year of sample from 1st August 2019 to 31st July 2020. Notably, the model assigns a sustained period of high volatility to consecutive days roughly coinciding with the onset of COVID-19 at the start of 2020. These first two months also coincide with the NBER recession indicator, which suggests the Markov-switching model is performing well.

Model smoothed probabilities show roughly 70% (30%) of days are classed as in the low (high) volatility state. Sustained periods of higher volatility is observed around key market events, such as around September 11 2001 and the collapse of Enron a month later, around the 2008-2009 GFC, and at the onset of COVID-19. Notably, these periods of sustained higher volatility align closely with the (Federal Reserve Bank of St. Louis, 2022) NBER recession indicator. Sustained times of calm are observed, such as in the leadup to the GFC, and after the Greek debt restructuring in 2012. The transition times between low and high states is fairly swift, and almost all transitions to the high state last for at least several consecutive days, suggesting system stability in both states. These results, along with the close fit of high volatility states coinciding with NBER recession periods, suggest robustness in our approach.

Low Regime

	Control	LM 1	LM 2	VA 1	VA 2	FB 1	FB 2	TR 1	TR 2	KE 1	KE 2
P_{t-1}	-	0.0173	0.0071	-0.0045	-0.0051	0.001**	0.001**	-0.0011	-0.001	-0.0121	-0.0093
N_{t-1}	-	-0.0347*	-0.0457**	-0.0109	-0.0142*	0.0005	0.0005	-0.0013	-0.0012	-0.0899**	-0.0836**
P_{t-2}	-	-	0.0261	-	0.0027	-	0	-	-0.0004	-	0.0051
N_{t-2}	-	-	0.036*	-	0.0106	-	0	-	-0.0003	-	-0.0345
R_{t-1}	-0.1324***	-0.15***	-0.1473***	-0.1382***	-0.1382***	-0.1637***	-0.1633***	-0.156***	-0.1555***	-0.1547***	-0.1561***
R_{t-2}	-0.0354**	-0.04**	-0.034*	-0.0374**	-0.0325*	-0.0375**	-0.0363*	-0.0395**	-0.0373*	-0.0363**	-0.0432**
VIX_{t-1}	-0.5634***	-0.5618***	-0.5377***	-0.5688***	-0.5575***	-0.5776***	-0.5752***	-0.5803***	-0.576***	-0.5763***	-0.5865***
VIX_{t-2}	0.8748***	0.8801***	0.8489***	0.8859***	0.8689***	0.8965***	0.8938***	0.9026***	0.8981***	0.881***	0.8918***
$Tedrate_{t-1}$	-0.0023	-0.0025	-0.0021	-0.0024	-0.0022	-0.0026	-0.0026	-0.0027	-0.0029	-0.0021	-0.0021
$Tedrate_{t-2}$	0.0002	0.0002	0.0001	0.0001	0	0.0005	0.0005	0.0006	0.0008	-0.0002	-0.0002
$Termspread_{t-1}$	-0.0046	-0.0051	-0.0053*	-0.0047	-0.0048	-0.005	-0.005	-0.0049	-0.0049	-0.005	-0.0049
$Termspread_{t-2}$	0.0041	0.0046	0.0048	0.0042	0.0043	0.0046	0.0045	0.0045	0.0045	0.0045	0.0045
$BAA10Y_{t-1}$	-0.0103***	-0.0097**	-0.0097**	-0.0101***	-0.01***	-0.0098	-0.0099	-0.0097**	-0.0096**	-0.01***	-0.0099***
$BAA10Y_{t-2}$	0.01***	0.0096**	0.0095**	0.0099***	0.0097**	0.0096**	0.0096**	0.0094**	0.0093**	0.0098***	0.0097
Intercept	-0.0019	-0.0016**	-0.002**	-0.001	-0.0015	-0.0023***	-0.0023***	-0.0032***	-0.0035**	-0.0012	-0.0011
σ^2	0.0***	0.0***	0.0***	0.0***	0.0***	0.0***	0.0***	0.0***	0.0***	0.0***	0.0***
R Squared	0.057	0.055	0.057	0.055	0.057	0.06	0.059	0.055	0.054	0.059	0.06
$\mathbb{P}(X_{t+1} \to Low)$	0.9843***	0.985***	0.9854***	0.9847***	0.9847***	0.9846***	0.9846***	0.9846***	0.9845***	0.9849***	0.9849***
AIC	-32988	-32989	-32985	-32984	-32979	-32994	-32986	-32990	-32985	-32992	-32985
BIC	-32817	-32792	-32762	-32788	-32756	-32797	-32764	-32794	-32763	-32795	-32762
n	3569	3599	3606	3587	3587	3572	3572	3595	3591	3589	3593

High Regime

	Control	LM 1	LM 2	VA 1	VA 2	FB 1	FB 2	TR 1	TR 2	KE 1	KE 2
P_{t-1}	-	0.0958	0.0679	-0.0001	0.001	0.005**	0.0048**	0.0058	0.0035	-0.3068	-0.3538
N_{t-1}	-	-0.1253*	-0.1392*	-0.0313	-0.0384	0.0028	0.0026	0.0037	0.0015	-0.358**	-0.3685**
P_{t-2}	-	-	0.0924	-	-0.0153	-	0.0009	-	0.0068*	-	0.2504
N_{t-2}	-	-	0.0466	-	0.0164	-	0.0008	-	0.0067*	-	0.0395
R_{t-1}	-0.0598	-0.0821*	-0.0834*	-0.0683	-0.0714	-0.1055**	-0.1056**	-0.1013**	-0.1021**	-0.0781*	-0.0792*
R_{t-2}	-0.0186	-0.0314	-0.032	-0.0237	-0.0175	-0.0294	-0.0275	-0.0345	-0.0285	-0.0229	-0.0234
VIX_{t-1}	0.8151***	0.8499***	0.8339***	0.8343***	0.8179***	0.772**	0.7676**	0.7882	0.7818	0.8202***	0.8081***
VIX_{t-2}	-0.5939	-0.6088**	-0.598	-0.5925*	-0.5893*	-0.562*	-0.5608*	-0.5659*	-0.5745*	-0.6034**	-0.5935*
$Tedrate_{t-1}$	0.0083	0.0087	0.0091	0.0081	0.0083	0.009	0.0092	0.0094	0.0101	0.0079	0.0081
$Tedrate_{t-2}$	-0.01	-0.0106	-0.0108	-0.0102	-0.0102	-0.0103	-0.0103	-0.011	-0.0113*	-0.0101	-0.0102
$Termspread_{t-1}$	0.0111	0.0102	0.0106	0.011	0.0108	0.0097	0.0095	0.0093	0.0085	0.0097	0.0098
$Termspread_{t-2}$	-0.0115	-0.0107	-0.0112	-0.0114	-0.0112	-0.0101	-0.0099	-0.0098	-0.009	-0.0102	-0.0102
$BAA10Y_{t-1}$	0.013	0.0137	0.0136	0.0136	0.0136	0.0141	0.0143	0.0145	0.0147	0.0131	0.0132
$BAA10Y_{t-2}$	-0.0127	-0.0133	-0.0132	-0.0133	-0.0133	-0.0137	-0.0138	-0.014	-0.0142	-0.0127	-0.0128
Intercept	-0.0071***	-0.0053*	-0.0068*	-0.0053	-0.0046	-0.0091***	-0.0097***	-0.0031	0.0012	-0.004*	-0.0048*
σ^2	0.0004***	0.0004***	0.0004***	0.0004***	0.0004***	0.0004***	0.0004***	0.0004***	0.0004***	0.0004***	0.0004***
R Squared	0.048	0.049	0.049	0.048	0.048	0.051	0.051	0.052	0.054	0.05	0.05
$\mathbb{P}(X_{t+1} \to Low)$	0.0344***	0.0334***	0.0328***	0.0339***	0.0339***	0.0337***	0.0337***	0.0342***	0.0342***	0.0336***	0.0338***
AIC	-32988	-32989	-32985	-32984	-32979	-32994	-32986	-32990	-32985	-32992	-32985
BIC	-32817	-32792	-32762	-32788	-32756	-32797	-32764	-32794	-32763	-32795	-32762
n	1592	1562	1555	1574	1574	1589	1589	1566	1570	1572	1568

Table 4: Markov regression results.
Results from a two-state Markov switching regression model on next day % returns of the S&P 500. Each model includes up to two days of lagged controls and sentiments, and is solved for each sentiment classifier separately. A control model is included for comparison. Variable coefficients and significance shown, $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

As with the simple OLS model, we observe generally $t-1$ negative sentiment is statistically significant, with a negative sign, consistently across both regimes. However, for the high regime we observe sentiment coefficient magnitudes are consistently roughly four times larger than for the low regime, regardless of the sentiment classifier used.[19] This increase in sentiment magnitude for the higher volatility state mirrors the results of Birru and Young (2022), and suggests sentiment effects are stronger during times of uncertainty. We also observe R-squared values are roughly double that of the simple OLS results, a result of the model performing effectively two grouped regressions over the data. The return variance of each regime, and regime transition rates are consistently highly significant, with much larger $Low \rightarrow Low$ transition rates. Together these results show the importance of taking into account that market conditions and market return predictability can vary greatly over time, and during different economic contexts.[20]

Overall, the inclusion of lagged sentiments improve R squared values across most models, even those where lagged sentiments are insignificant. Generally, the simple robust dictionary methods of LM and KE perform the best, followed by FB. Similar patterns as for OLS table are observed, with N_{t-1} sentiment significant for LM and KE, and P_{t-1} for FB. Interestingly, the KE 2 model performs the best, with the greatest R-squared improvement across both regimes, followed by FB and then LM. This speaks to the significance and importance of the words identified by Ke et al. (2019). Interestingly, despite mostly improved R-squared values across all models and both regimes, generally neither AIC nor BIC values improve. This indicates that although sentiment improves model accuracy, much of the information content is being subsumed from the control variables.

Note while we do not identify the identical levels of sentiment significance as Birru and Young (2022), this is due to several reasons. First, our study period is roughly three times longer, and covers chaotic times outside their study period, including September 11 and the onset of CVOID-19. Second, their classification of low (high) regimes of uncertainty for each day is based on the VIX, while in contrast our low (high) volatility regime are endogenously determined by the Markov model. Finally, their sentiment measure is based on the FEARS index by Da et al. (2015), while ours is derived from aggregate news publications, and so may differ. In addition, our regressions contain additional controls which are highly likely to hold significance, including lagged daily returns and

[19] As an aside, the sentiment coefficients of the OLS regression results Table 3.1 lie between the coefficients of the low and high regime states, which is what we would expect.

[20] As a side note, lagged returns are most significant in the low volatility state. However, for the high volatility state VIX_{t-1} is primarily significant. The sign of VIX also changes to positive for the high regime, and is larger in magnitude by roughly 50%. This demonstrates heteroskedasticity is present. Days with the largest swings in market returns coincide with the high regime, see Figure 2.

VIX, the inclusion of which would only serve to reduce the significance of our sentiment measures compared to their results. The fact our lagged sentiment measures are still significant despite controlling for both lagged returns, VIX, return volatility, and across both regimes, speaks to the information content and strength of news sentiment.

Our results paint an interesting picture. Consistent with the simpler OLS model, regime results suggest that negative news has more predictive power than positive news, and that sentiment is stronger in times of uncertainty, however in itself may not always capture new information beyond existing volatility measures, as this information may already be accounted for directly through market volatility, or proxies such as the VIX.

3.3 GARCH

GARCH model results are detailed in Table 5. Consistent with prior results, negative sentiments are the most statistically significant, and dominate across all models, except for FB.

	Control	LM 1	LM 2	VA 1	VA 2	FB 1	FB 2	TR 1	TR 2	KE 1	KE 2
P_{t-1}	-	0.0378	0.0193	-0.0052	-0.0066	0.0011***	0.0011**	-0.0003	-0.0004	-0.0684	-0.0954
N_{t-1}	-	-0.0184	-0.0341*	-0.0099*	-0.0158**	0.0006**	0.0006**	-0.0006	-0.0006	-0.0882***	-0.0829**
P_{t-2}	-	-	0.0384	-	0.006	-	0	-	0.0004	-	0.0878
N_{t-2}	-	-	0.0345**	-	0.0136**	-	0	-	0.0004	-	-0.018
R_{t-1}	-0.0897***	-0.1037***	-0.1065***	-0.0925***	-0.0983***	-0.1107***	-0.1102***	-0.1086***	-0.1091***	-0.1092***	-0.1073***
R_{t-2}	-0.0235	-0.0307*	-0.0293*	-0.0246	-0.0222	-0.0278*	-0.0296*	-0.0302*	-0.0288*	-0.0276*	-0.0322**
VIX_{t-1}	-0.1781	-0.1772	-0.1881	-0.1755	-0.193	-0.192	-0.1886	-0.1941	-0.1967	-0.2069	-0.2027
VIX_{t-2}	0.2691	0.2788*	0.2891*	0.2746*	0.284*	0.2891*	0.2863*	0.2925*	0.2937*	0.2949*	0.2929*
$Tedrate_{t-1}$	0.0016	0.0017	0.0021	0.0016	0.0018	0.0017	0.0017	0.0016	0.0017	0.0019	0.0018
$Tedrate_{t-2}$	-0.0028	-0.0029	-0.0032	-0.0031	-0.0031	-0.0028	-0.0028	-0.0028	-0.0029	-0.0033	-0.0032
$Termspread_{t-1}$	-0.002	-0.0025	-0.0025	-0.0022	-0.002	-0.0023	-0.0023	-0.0024	-0.0023	-0.0024	-0.0024
$Termspread_{t-2}$	0.0017	0.0022	0.0022	0.0019	0.0017	0.002	0.0021	0.0021	0.0021	0.0021	0.0021
$BAA10Y_{t-1}$	-0.0105**	-0.0098**	-0.0099**	-0.0103**	-0.0101**	-0.0101**	-0.0101**	-0.0098**	-0.0099**	-0.01**	-0.0098**
$BAA10Y_{t-2}$	0.0103**	0.0096**	0.0096**	0.0102**	0.0099**	0.0099**	0.0099**	0.0097**	0.0098**	0.0098**	0.0097**
Intercept	0.0004	0.0002	-0.0003	0.0013	0.0005	-0.0002	-0.0001	-0.0002	0	0.0011*	0.001*
ω	0***	0***	0***	0***	0***	0***	0***	0***	0***	0***	0***
α_{t-1}	0.1244***	0.1241***	0.1235***	0.1243***	0.1242***	0.1245***	0.1243***	0.1244***	0.1242***	0.124***	0.1242***
β_{t-1}	0.8606***	0.8613***	0.8622***	0.861***	0.861***	0.8608***	0.861***	0.8608***	0.861***	0.8614***	0.8611***
Adj. R Squared	0.0088	0.0109	0.0118	0.0097	0.0103	0.0118	0.0117	0.0109	0.0110	0.0118	0.0116
AIC	-6.4648	-6.4649	-6.4651	-6.4646	-6.4647	-6.4656	-6.4648	-6.4647	-6.4639	-6.4655	-6.4651
BIC	-6.447	-6.4446	-6.4422	-6.4443	-6.4418	-6.4453	-6.442	-6.4444	-6.4411	-6.4452	-6.4422
n						5,161					

Table 5: GARCH regression results.
Results for our GARCH $(1, 1)$ model, where α and β are the coefficients of lagged error and variance terms respectively. Adjusted R Squared, AIC, and BIC values are different to other table results due to the different model specification of GARCH. A control regression is included for comparison. Robust standard errors used. $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Though sentiment coefficients are roughly 1/3 to 1/2 the size of coefficients from OLS results Table 3, the same general level of significance is observed. Results also show a moderate to strong level of volatility persistence $\beta_1 \approx 0.86$, though variance decays for all models, $\alpha + \beta < 1$. Also, the return variance coefficient ω is zero and significant across all models, similar to the low values identified from the Markov-switching model. Notably, α and β are significant at the 1% level across all models, reflecting heteroskedasticity is present.

Overall, the reduced sentiment coefficient magnitudes indicate that news sentiments capture some part of the heteroskedasticity of market returns. This is further supported by the fact that although adjusted R squared values improve over the control model, BIC values do not improve.[21]

3.4 VAR

Results of the vector autoregression model with lag order 2 are detailed in Table 6. For simplicity, and to aid in interpretability, the data is presented with model columns grouped by dependent variable of interest, from left to right: S&P 500 % returns, positive sentiment, and negative sentiment. Each classifier is a separate VAR model, which solves a vector of equations with each variable as the response, per Equation 6. Hence all LM columns correspond to results from the same VAR_{LM} model, and so on.[22]

	S&P 500 % Returns					Positive Sentiment					Negative Sentiment				
	LM	VA	FB	TR	Ke	LM	VA	FB	TR	Ke	LM	VA	FB	TR	Ke
P_{t-1}	0.088*	0.007	0.002	-0.003	-0.152	0.289***	0.197***	0.095	0.187	0.234***	0.08***	0.034**	0.274***	0.179	-0.002
N_{t-1}	-0.062**	-0.027**	0.001	-0.003*	-0.165***	0.053***	-0.022	0.051	0.051	0.017***	0.282***	0.35***	0.256***	0.3**	0.161***
P_{t-2}	-0.013	-0.007	0	0.001	0.12	0.214***	0.142***	0.039	0.353***	0.153***	0.076***	0.036**	0.283***	-0.12	0
N_{t-2}	0.013	0.007	0	0.001	0.007	0.035***	0.004	0.021	0.232**	-0.003	0.192***	0.25***	0.246***	-0.006	0.132***
R_{t-1}	-0.087***	-0.079***	-0.097***	-0.101***	-0.083***	0.028***	0.095***	1.321	3.355**	-0.003	-0.034**	-0.069**	-1.525	-3.824**	0.007
R_{t-2}	-0.027*	-0.023	-0.02	-0.025	-0.025	-0.015***	-0.007	-0.339	-2.276**	-0.004**	0.016*	0.076***	0.005	2.408**	0.019***
VIX_{t-1}	0.472***	0.465***	0.43**	0.447***	0.433***	0.084	0.221	-2.159	1.11	-0.025	-0.059	0.037	6.79	-3.296	-0.079*
VIX_{t-2}	-0.398**	-0.394**	-0.368**	-0.367**	-0.379**	-0.131**	-0.21	-2.036	-10.05	0.027	0.133	0.173	-0.927	14.274	0.051
$Tedrate_{t-1}$	0.007**	0.007**	0.007**	0.007**	0.007**	0.001	0.001	0.083	0.086	-0.001	0	-0.001	-0.228	-0.104	-0.001
$Tedrate_{t-2}$	-0.009**	-0.009**	-0.009**	-0.009**	-0.009**	-0.002*	0	-0.159	-0.151	0	-0.002	-0.005	0.224	0.147	0.001
$Termspread_{t-1}$	0.004	0.004	0.004	0.004	0.004	0.001	0.005	0.107	0.247	0.001***	-0.003	-0.006	-0.161	-0.242	-0.001
$Termspread_{t-2}$	-0.004	-0.005	-0.004	-0.004	-0.004	-0.001	-0.005	-0.09	-0.198	-0.001***	0.002	0.005	0.138	0.192	0.001
$BAA10Y_{t-1}$	0.005	0.005	0.005	0.005	0.004	-0.002	-0.018**	-0.761*	-0.41	0.001	0.009***	0.02**	0.984**	0.418	0.001
$BAA10Y_{t-2}$	-0.005	-0.005	-0.005	-0.005	-0.004	0.002	0.015**	0.703*	0.367	-0.001	-0.008**	-0.017**	-0.909*	-0.378	-0.001
Intercept	0	0.001	-0.001	-0.002	0.001	0.004***	0.065***	-0.183**	0.302**	0.002***	0.007***	0.01***	0.615***	-0.723***	0.004***
R Squared	0.0212	0.0204	0.0209	0.0217	0.0209	0.2416	0.1398	0.0199	0.0873	0.1257	0.2386	0.3457	0.0203	0.0864	0.0687
Log likelihood	112,199	100,706	67,565	71,589	120,984	112,199	100,706	67,565	71,589	120,984	112,199	100,706	67,565	71,589	120,984
AIC	-63.32	-58.87	-46.08	-47.57	-66.73	-63.32	-58.87	-46.08	-47.57	-66.73	-63.32	-58.87	-46.08	-47.57	-66.73
BIC	-63.19	-58.73	-45.88	-47.44	-66.59	-63.19	-58.73	-45.88	-47.44	-66.59	-63.19	-58.73	-45.88	-47.44	-66.59
n			5,159					5,159					5,159		

Table 6: Vector autoregression results.
This table details VAR analysis results for models using sentiments from each classifier separately, from VAR_{LM} to VAR_{KE}. For ease of comparison, results for each VAR regression model is grouped based on the equation response variable of interest, in order: S&P 500 returns (left), positive sentiment (middle), and negative sentiment (right). The R squared value of each equation is shown, along with the model log likelihoods and AIC and BIC values of each VAR model. Coefficient significance is also shown: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Results show strong autocorrelation effects across both market returns, and both positive and negative sentiments. For equations with market returns as the response, we observe similar coefficient magnitudes and significance to those of prior models. R squared values are generally consistent across classifiers, with N_{t-1} sentiment most significant. Overall KE and LM have the highest model log likelihood, R squared, and BIC values, reflecting the robustness of dictionary classifiers.

[21] Note that in addition to the GARCH models shown here, we also briefly considered multi-frequency GarchMIDAS model specifications, which allowed for simultaneous multi-frequency heteroskedasticity modelling. Such a model could be useful, as longer-patterns in news sentiments on different frequencies could capture different information (Hendershott et al., 2015). However, these models had several limitations as they did not allow for the inclusion of exogenous variable controls. In addition, they suffered from convergence issues. When they did, results were mixed and challenging to interpret, suggesting inaccurate model specifications. For these reasons, GarchMIDAS models were not explored further. Instead, we opted for a wavelet analysis to investigate sentiment effects at different frequencies over time. See Section 3.6.

[22] Results for the remaining control variables as responses in the VAR analysis are excluded for brevity.

Interestingly however, for equations of positive and negative sentiments as response variables, almost all lagged sentiment terms are highly significant, with all coefficients positive and greatest for the same predictor-response pairs, i.e. positive-positive and negative-negative, reflecting good and bad news have strong autocorrelation effects. This indicates any increase in news sentiments over the previous two days increases future news sentiments, regardless if the prior news is positive or negative, indicating news cycles and coverage is persistent over at least the last two days, with diminishing effects over time.[23]

Overall these results provide strong evidence that news coverage, sentiment, and volume does not strictly reflect, nor follow market returns, but instead are in-part self driven, suggesting news markets operate as a separate information system. Though news sentiments may capture investor sentiments and may lead next day returns, the strongest indicator of future news sentiments is the past strength and volume of news sentiments (Bybee et al., 2020).

3.5 Markov-switching VAR

Results for the Markov-switching VAR (MS-VAR) model are presented below. Plots of smoothed probabilities for the LM model being in a high volatility state is shown in Figure 8, while smoothed probabilities for the remaining four classifiers are shown in Appendix A.1 for brevity. Impulse response functions IRF's for all models and across both regimes are shown in Figure 9. For simplicity, returns, positive, and negative sentiments are shown, with blue (red) lines representing IRF's for the low (high) market volatility regimes. Full plots for each classifier are provided in Appendix A.2.

[23] Another aspect worth noting is that for KE, for negative news as the response variable, only prior negative news sentiments are significant, with both having positive coefficients, showing negative new perpetuates further negative news.

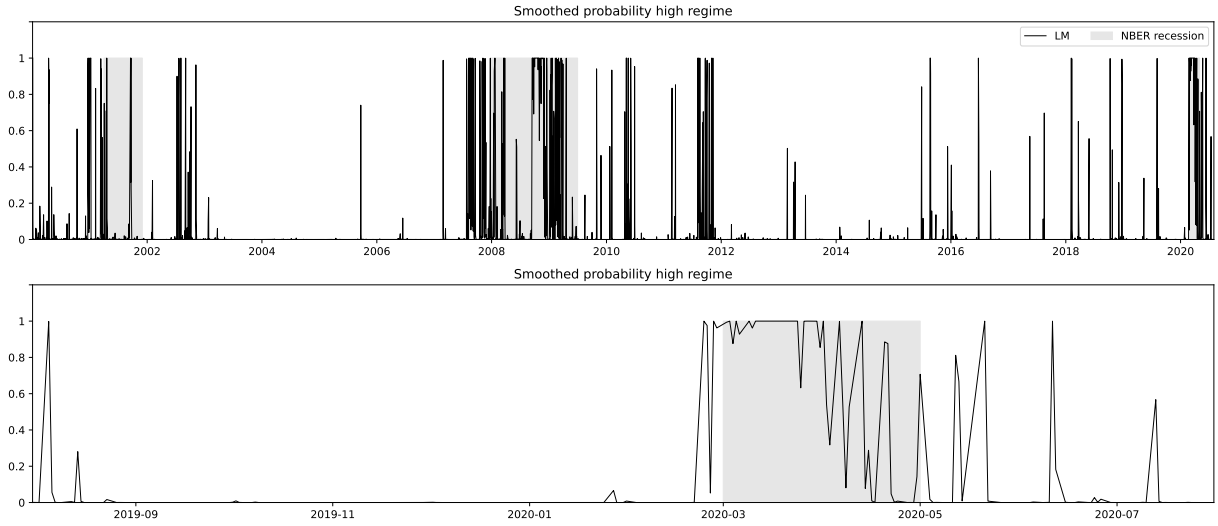


Figure 8: Plots of smoothed probabilities of the S&P 500 being in a high volatility state over time for the LM MS-VAR model. The two plots show the full 20 year sample period (top) and last year of sample (bottom), with switching rates: $\mathbb{P}(X_{Low,t+1} \rightarrow Low) = 0.985$ and $\mathbb{P}(X_{High,t+1} \rightarrow Low) = 0.266$. The plots mirror those of the simpler Markov-switching OLS regression Figure 6, with similar regime switching rates, and sustained periods of high volatility aligning closely with the NBER recession indicator around periods of market instability, such as the 2008-2009 GFC, and the start of 2020 with COVID-19.

Overall Figure 8 closely matches prior results of the simpler Markov-switching OLS results Figure 6, and though the switching rate from high to low regime is greater, they are generally closely aligned with very similar behaviour, indicating the model is performing well.[24] Note that despite switching rates and low (high) volatility identification closely matching prior simpler model results, it was somewhat challenging to get the MS-VAR models to converge, and considerable iteration and fine-tuning of hyperparameters was needed. However, once a fine balance was found, all sentiment models converged. Model instability was almost certainly due to the large number of variables included, since AR models with so many endogenous inputs are not necessarily well-defined, and hence may not be stable. Still, the fact we achieve similar results consistent with prior MS-OLS results supports our findings and modelling procedure.

[24] Similar behaviour is observed across all other sentiment models, see Appendix A.1.

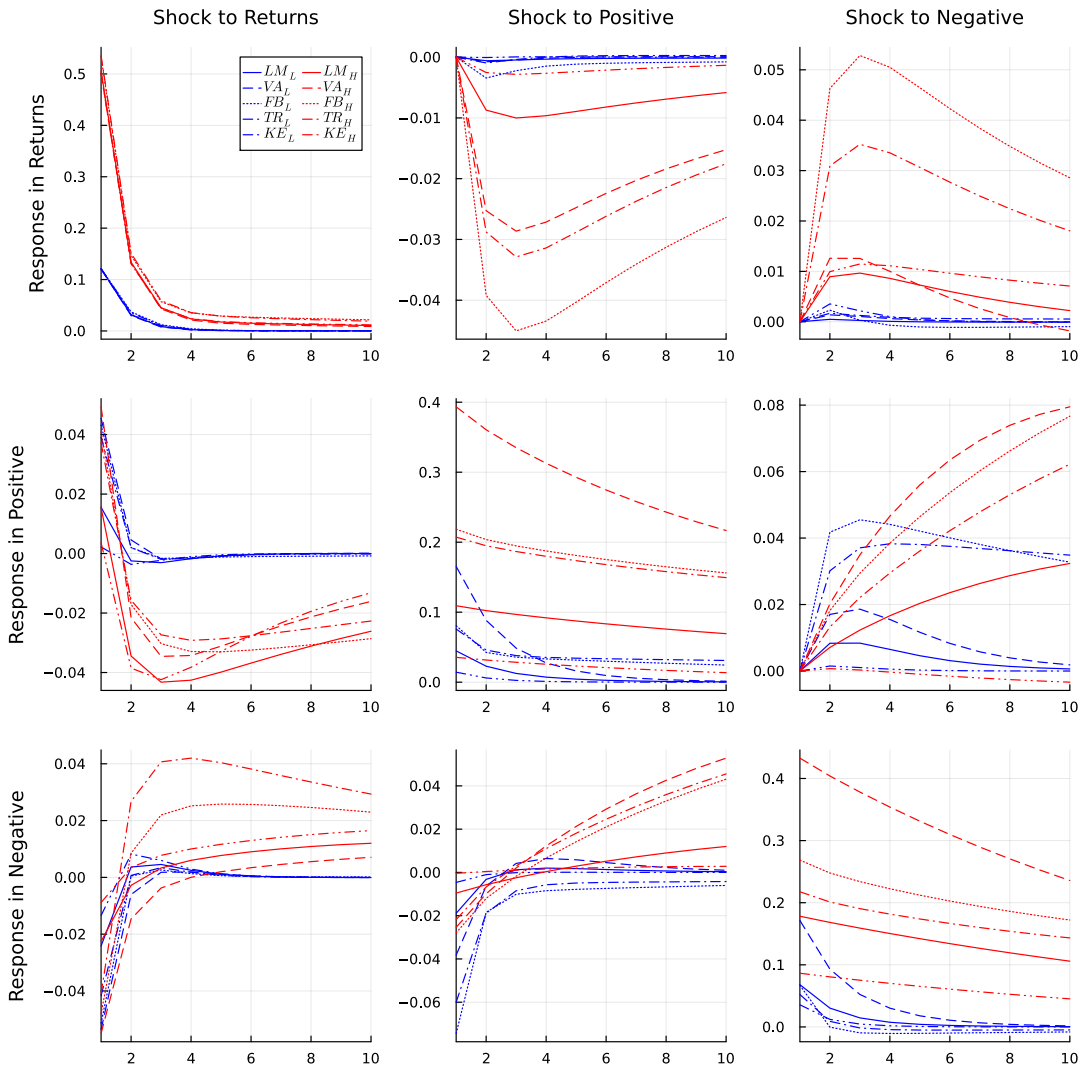


Figure 9: Plots showing responses over the next ten days following a unit shock in returns, positive news sentiment, and negative news sentiment. Responses are shown for both low (blue), and high (red) market volatility states. All sentiment models are included, and though each MS-VAR model includes all additional controls in the modelling procedure, only returns and sentiments are shown here. See Appendix A.2 for full plots. We observe responses are consistently greater when market volatility is higher. In particular, we observe that sentiments, both $P \to P$ and $N \to N$ exhibit lasting persistence when markets are most volatile. Returns shocks also have lasting effects particularly in the high regime. While shocks to returns (news) have some effects on news (returns), these effects are an order of magnitude smaller than news shocks have on future news, particularly when markets are more volatile. This suggests news and news markets are largely self-driven, particularly when markets are most volatile. These results mirror of VAR results, see Section 3.4.

IRF results show key differences between low and high regimes, with responses consistently higher when markets are more volatile. During these times, returns are particularly susceptible to market shocks, and though news shocks impacts market returns, the effect is an order of magnitude less than for a shock in returns. Also, while shocks to returns have initial positive (negative) impacts on positive (negative) sentiments, in the high volatility state these initial responses are uniquely followed by reversals over the following two weeks, with positive news having larger reversals by

roughly a factor of two. Hence momentum reversal patterns are not limited to returns, but are also observed in news cycles as well, and that contemporaneous news reports are relatively short-lived. In addition, contemporaneous negative news shocks have stronger effects on future news than positive news shocks. These effects are strongest when market volatility is highest, with long-lasting effects observed. On close inspection, negative sentiment responses to negative sentiments are larger than positive news responses to positive news shocks by up to a factor of two, see for example LM.[25]

Overall, results indicate that return shocks consistently drive news more than news shocks drive returns. In addition, the effects of both positive and negative news shocks on future news are far stronger and longer lasting when markets are more volatile. Our results indicate news markets and stock markets operate as two somewhat separate information system, with contagion effects transmitted between them. Feedback loops caused by reader interest and excitement around particular news topics or themes likely increases news volume and the strength of news sentiments, exacerbating market inefficiency via incorrect shared beliefs. We would then expect downward pressure on stock markets to be particularly notable when markets are most volatile, and may expect reversals of short-term news overreactions (Hirshleifer, 2020; Mullainathan and Shleifer, 2005; Bybee et al., 2020). The fact we observe news reversals from returns shocks, return reversals in the days ex-post of positive and negative news shocks, and long-lasting news autocorrelation effects particularly in the high regime gives credence to this theory, and highlights the significance of the social transmission channel (Hirshleifer, 2020).

3.6 Wavelet

Results from the wavelet analysis Equation 7 are detailed below. For each sentiment classifier, we calculate the coherence between the daily percentage change in S&P 500 market returns and both positive and negative news sentiments separately. Results for the LM classifier are shown in Figure 10, with results for the other five sentiment classifiers detailed in Appendix A.3.

[25] The exception is VA, which is roughly the same.

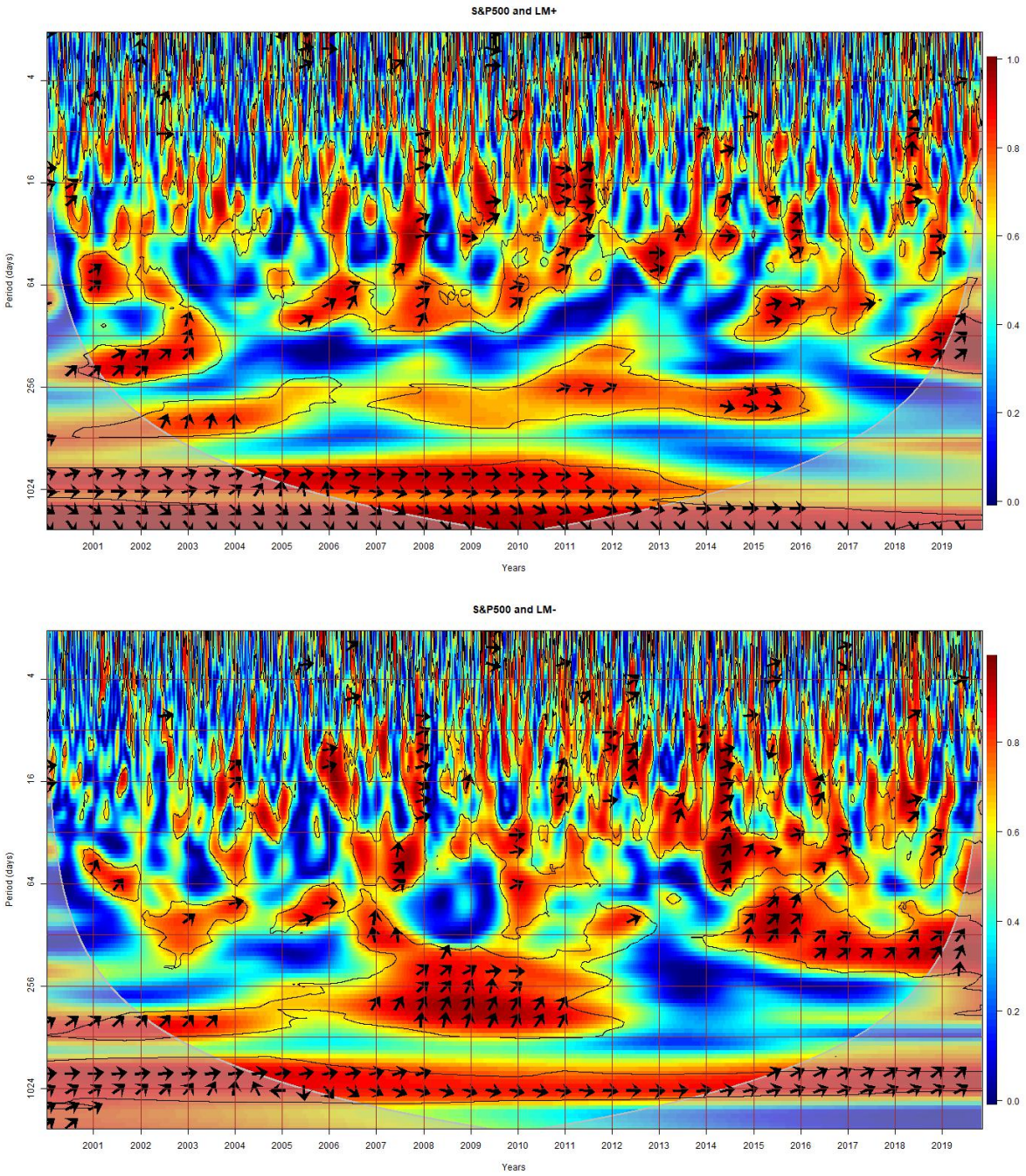


Figure 10: Wavelet coherence results between daily S&P 500 returns and daily market news sentiments of LM, positive (top) and negative (bottom). Values of 95% or greater are shown within the red areas of the black contours. Phase arrows represent: in-phase (right), anti-phase (left), news sentiment leading S&P 500 (up), S&P 500 leading news sentiment (down). The cone of influence is shown in grey, outside of which results may be affected by boundary condition effects.

Across both positive and negative sentiments for periods shorter than roughly two to three weeks, we observe frequent switching between both very high, and low coherence. This rapid switching is not surprising, as news sentiments can oscillate quickly and change day-to-day. For longer periods of between roughly 1 to 6 months we observe much stronger persistence, and groupings of years with particularly high coherence. For example, particularly notable red-hot vertical sections of high coherence between periods of roughly two weeks to a year are observed. The first of these corresponds to the 2008-2009 GFC, while the second roughly coincides with the European Greek Debt crisis around 2011. Finally, at periods of roughly 1 year, we observe large and persistent sections of high coherence. The start of the first section roughly coincides with the collapse of Enron in 2001, and lasts for several years after. The second notable large patch starts at roughly the beginning of 2007, and also subsequently persists for roughly the next 8 years up until 2015.

Our results reflect those of Hong et al. (2000) of gradual information diffusion over low monthly frequencies, as we observe a similar pattern of slow negative information diffusion that occurs over time. Further, the up-right direction of arrows indicate that news sentiments are both in-phase and lead aggregate market returns. Notably, our wavelet analysis suggests evidence of persistent negative sentiment effects at lower frequencies of several months, or longer. We observe negative sentiment effects are stronger, particularly around times of market downturns, and highest uncertainty. However for higher frequencies, news is unlikely to have a persistent effect, with frequent switching between high and low coherence. These results are in-agreement with prior literature, which suggests that long-running sentiment effects are persistent (Garcia, 2013), however short-term sentiments typically predict momentum reversals (Birru and Young, 2022).

4 Conclusion

Using a large proprietary news dataset, we derive measures of daily positive and negative sentiments using both traditional lexicon-based methods, and modern neural networks over a 20 year window. By incorporating these sentiments in a barrage of different regression models we investigate different channels through which sentiments conveyed through news may forecast future returns of the S&P 500. Throughout we account for potential asymmetric positive and negative effects from sentiment, and we explore several distinct mechanisms through which news sentiment may act.

We demonstrate three main findings. First, in agreement with prior literature, we find evidence overreaction to negative news dominates reactions to positive news, with these effects strongest when market volatility is greatest. Second, we show heteroskedasticity effects overlap with observed news sentiments and return reversals, and that accounting for heteroskedasticity partially subsumes sentiment effects by up to 50%. Finally, our results from VAR, Markov-switching VAR, and wavelets analysis emphasises that news markets and stock markets as separate information systems that, while often overlap, also convey information that can influence each other. Unique shocks within both systems can influence returns and future sentiments, and we document similar momentum reversal patterns in both the news market and stock market, particularly when volatility is highest. This is important, since during these times the market for news and news demand is the primary driver of future news sentiments, and this in-turn then contributes to market inefficiency through the social transmission channel.

Our results highlight the information content of news text, and demonstrate its importance as an information source. We substantiate the robustness of dictionary approaches as, while they may not be able to take into account semantic meaning, they are still able to capture important information without the computational expense of more advanced algorithms like neural networks and large language models. Most importantly, we provide evidence sudden, and long-lasting news sentiment feedback effects are strongest when markets are most volatile, and this can exacerbate market volatility through intensified downward pressure on financial markets.

References

Abreu, D. and Brunnermeier, M. K. (2003). "Bubbles and Crashes". *Econometrica*, 71.1, pp. 173–204.

Akhtaruzzaman, M., Boubaker, S., and Sensoy, A. (2021). "Financial contagion during COVID–19 crisis". *Finance Research Letters*, 38, p. 101604.

Allen, F. and Gale, D. (2000). "Financial Contagion". *Journal of Political Economy*, 108.1, pp. 1–33.

American Association of Individual Investors (2022). *Sentiment Survey Historical Data*. https://www.aaii.com/sentimentsurvey/sent_results. Accessed: 12 March 2022.

Araci, D. (2019). "FinBERT: Financial Sentiment Analysis with Pre-trained Language Models". *arXiv preprint arXiv:1908.10063*.

Baker, M. and Wurgler, J. (2006). "Investor Sentiment and the Cross-Section of Stock Returns". *The Journal of Finance*, 61.4, pp. 1645–1680.

Barbieri, F., Camacho-Collados, J., Neves, L., and Espinosa-Anke, L. (2020). "TweetEval: Unified Benchmark and Comparative Evaluation for Tweet Classification". *arXiv preprint arXiv:2010.12421*.

Baumeister, R. F., Bratslavsky, E., Finkenauer, C., and Vohs, K. D. (2001). "Bad is Stronger than Good". *Review of General Psychology*, 5.4, pp. 323–370.

Binsbergen, J. H. van, Bryzgalova, S., Mukhopadhyay, M., and Sharma, V. (2024). *(Almost) 200 Years of News-Based Economic Sentiment*. Tech. rep. National Bureau of Economic Research.

Birru, J. and Young, T. (2022). "Sentiment and uncertainty". *Journal of Financial Economics*, 146.3, pp. 1148–1169.

Bybee, L., Kelly, B., Manela, A., and Xiu, D. (2024). "Business News and Business Cycles". *The Journal of Finance*, 79.5, pp. 3105–3147.

Bybee, L., Kelly, B. T., Manela, A., and Xiu, D. (2020). *The Structure of Economic News*. Tech. rep. National Bureau of Economic Research.

Chen, H., De, P., Hu, Y., and Hwang, B.-H. (2014). "Wisdom of Crowds: The Value of Stock Opinions Transmitted Through Social Media". *The Review of Financial Studies*, 27.5, pp. 1367–1403.

Chen, Y., Kelly, B. T., and Xiu, D. (2022). "Expected Returns and Large Language Models". *Available at SSRN 4416687*.

Da, Z., Engelberg, J., and Gao, P. (2011). "In Search of Attention". *The Journal of Finance*, 66.5, pp. 1461–1499.

Da, Z., Engelberg, J., and Gao, P. (2015). "The Sum of All FEARS Investor Sentiment and Asset Prices". *The Review of Financial Studies*, 28.1, pp. 1–32.

Daniel, K., Hirshleifer, D., and Sun, L. (2020). "Short-and Long-Horizon Behavioral Factors". *The Review of Financial Studies*, 33.4, pp. 1673–1736.

De Long, J. B., Shleifer, A., Summers, L. H., and Waldmann, R. J. (1990). "Noise Trader Risk in Financial Markets". *Journal of Political Economy*, 98.4, pp. 703–738.

Devlin, J. (2018). "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding". *arXiv preprint arXiv:1810.04805*.

Dewandaru, G., Masih, R., and Masih, A. M. M. (2016). "Contagion and interdependence across Asia-Pacific equity markets: An analysis based on multi-horizon discrete and continuous wavelet transformations". *International Review of Economics and Finance*, 43, pp. 363–377.

Engelberg, J. E. and Parsons, C. A. (2011). "The Causal Impact of Media in Financial Markets". *The Journal of Finance*, 66.1, pp. 67–97.

Engle, R. F. and Bollerslev, T. (1986). "Modelling the persistence of conditional variances". *Econometric Reviews*, 5.1, pp. 1–50.

Federal Reserve Bank of St. Louis (2022). *FRED Economic Data*. https://fred.stlouisfed.org/. Accessed: 13 March 2022.

Gallegati, M. (2012). "A wavelet-based approach to test for financial market contagion". *Computational Statistics & Data Analysis*, 56.11, pp. 3491–3497.

Gao, Z., Ren, H., and Zhang, B. (2020). "Googling Investor Sentiment around the World". *Journal of Financial and Quantitative Analysis*, 55.2, pp. 549–580.

Garcia, D. (2013). "Sentiment during Recessions". *The Journal of Finance*, 68.3, pp. 1267–1300.

Garcia, D., Hu, X., and Rohrer, M. (2023). "The colour of finance words". *Journal of Financial Economics*, 147.3, pp. 525–549.

Gentzkow, M., Kelly, B., and Taddy, M. (2019). "Text as Data". *Journal of Economic Literature*, 57.3, pp. 535–574.

Hendershott, T., Livdan, D., and Schürhoff, N. (2015). "Are institutions informed about news?" *Journal of Financial Economics*, 117.2, pp. 249–287.

Heston, S. L. and Sinha, N. R. (2017). "News vs. Sentiment: Predicting Stock Returns from News Stories". *Financial Analysts Journal*, 73.3, pp. 67–83.

Hirshleifer, D. (2020). "Presidential Address: Social Transmission Bias in Economics and Finance". *The Journal of Finance*, 75.4, pp. 1779–1831.

Hong, H., Lim, T., and Stein, J. C. (2000). "Bad News Travels Slowly: Size, Analyst Coverage, and the Profitability of Momentum Strategies". *The Journal of Finance*, 55.1, pp. 265–295.

Hutto, C. and Gilbert, E. (2014). "VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text". *Proceedings of the International AAAI Conference on Web and Social Media*. Vol. 8. 1, pp. 216–225.

Ke, Z. T., Kelly, B. T., and Xiu, D. (2019). *Predicting Returns With Text Data*. Tech. rep. National Bureau of Economic Research.

Loughran, T. and McDonald, B. (2011). "When Is a Liability Not a Liability? Textual Analysis, Dictionaries, and 10-Ks". *The Journal of Finance*, 66.1, pp. 35–65.

Minsky, H. P. (1977). "The Financial Instability Hypothesis: An Interpretation of Keynes and an Alternative to"Standard" Theory". *Challenge*, 20.1, pp. 20–27.

Mullainathan, S. and Shleifer, A. (2005). "The Market for News". *American Economic Review*, 95.4, pp. 1031–1053.

Obaid, K. and Pukthuanthong, K. (2022). "A picture is worth a thousand words: Measuring investor sentiment by combining machine learning and photos from news". *Journal of Financial Economics*, 144.1, pp. 273–297.

Refinitiv (2021). *Refinitiv Eikon with DataStream.* Available at: Subscription Service. Accessed: 12 March 2022.

Shapiro, A. H., Sudhof, M., and Wilson, D. J. (2022). "Measuring news sentiment". *Journal of Econometrics*, 228.2, pp. 221–243.

Tetlock, P. C. (2007). "Giving Content to Investor Sentiment: The Role of Media in the Stock Market". *The Journal of Finance*, 62.3, pp. 1139–1168.

Tetlock, P. C. (2010). "Does Public Financial News Resolve Asymmetric Information?" *The Review of Financial Studies*, 23.9, pp. 3520–3557.

Tetlock, P. C., Saar-Tsechansky, M., and Macskassy, S. (2008). "More Than Words: Quantifying Language to Measure Firms' Fundamentals". *The Journal of Finance*, 63.3, pp. 1437–1467.

Vaswani, A. (2017). "Attention is All You Need". *Advances in Neural Information Processing Systems.*

Yang, Y., UY, M. C. S., and Huang, A. (2020). "FinBERT: A Pretrained Language Model for Financial Communications". *arXiv preprint arXiv:2006.08097.*

Yu, J. and Yuan, Y. (2011). "Investor sentiment and the mean–variance relation". *Journal of Financial Economics*, 100.2, pp. 367–381.

Zhang, X. F. (2006). "Information Uncertainty and Analyst Forecast Behavior". *Contemporary Accounting Research*, 23.2, pp. 565–590.

A Appendix

A.1 MS-VAR Smoothed Probabilities

Plots of smoothed probabilities from the Markov-switching VAR analysis, showing the probability each day is classed as in a high volatility state. Each plot shows the full sample (top), and last year of sample (bottom). The four sentiments used in each model are: VA, KE, FB, and TR, while LM is presented in Section 3.5.

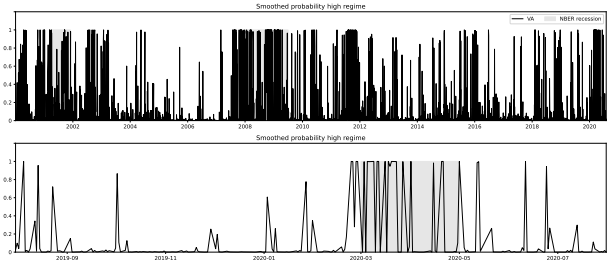


Figure 11: MS-VAR smoothed probabilities, VA sentiment.

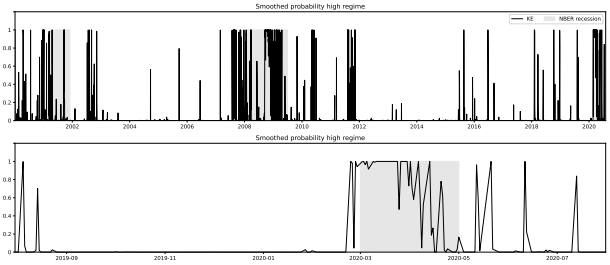


Figure 12: MS-VAR smoothed probabilities, KE sentiment.

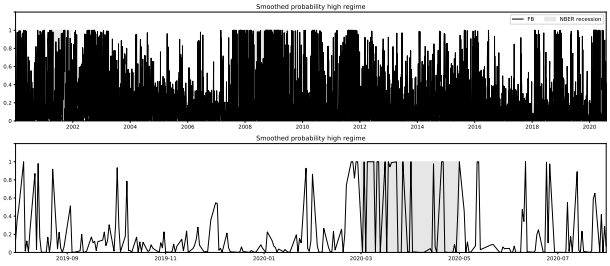


Figure 13: MS-VAR smoothed probabilities, FB sentiment.

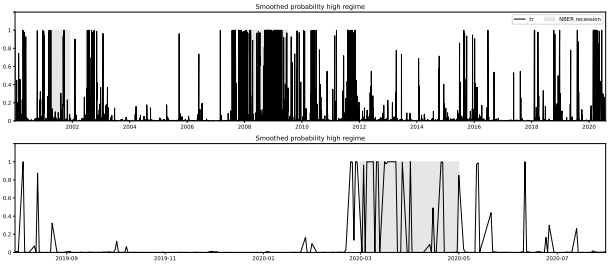


Figure 14: MS-VAR smoothed probabilities, TR sentiment.

A.2 MS-VAR IRF Plots

Full IRF plots for all five classifiers: LM, VA, FB, TR, KE from the Markov-switching VAR analysis. Each Figure is a 7×7 grid of IRF plots, as there are seven variables total. The low (high) market volatility regime is indicated in black (red). The vertical axis of plots are fixed according to the maximum ranges per row.

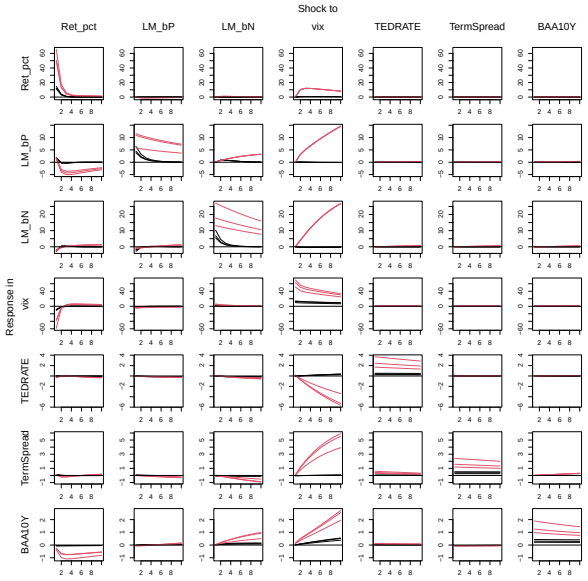


Figure 15: Full LM IRF plots.

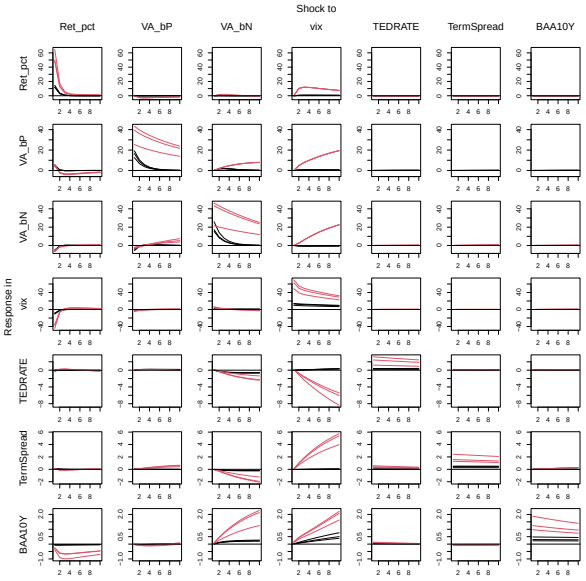


Figure 16: Full VA IRF plots.

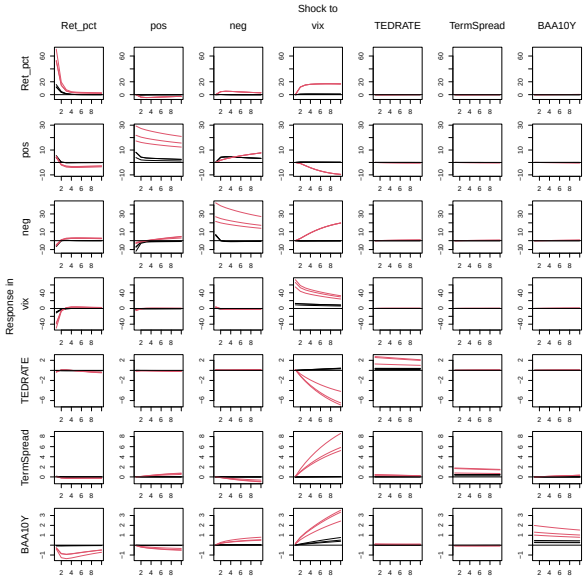


Figure 17: Full FB IRF plots.

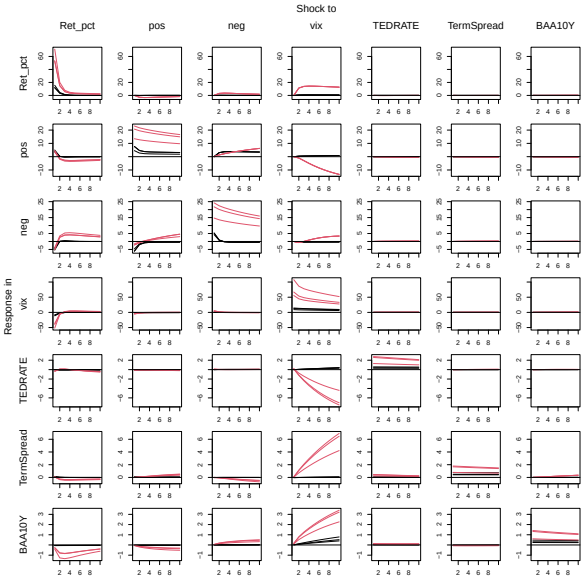


Figure 18: Full TR IRF plots.

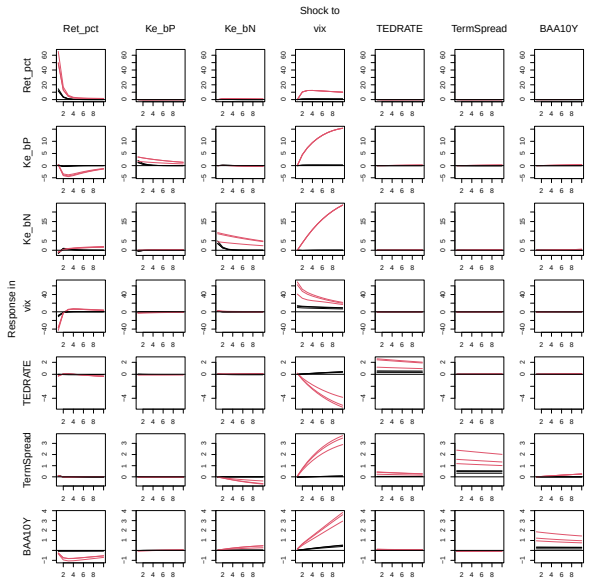
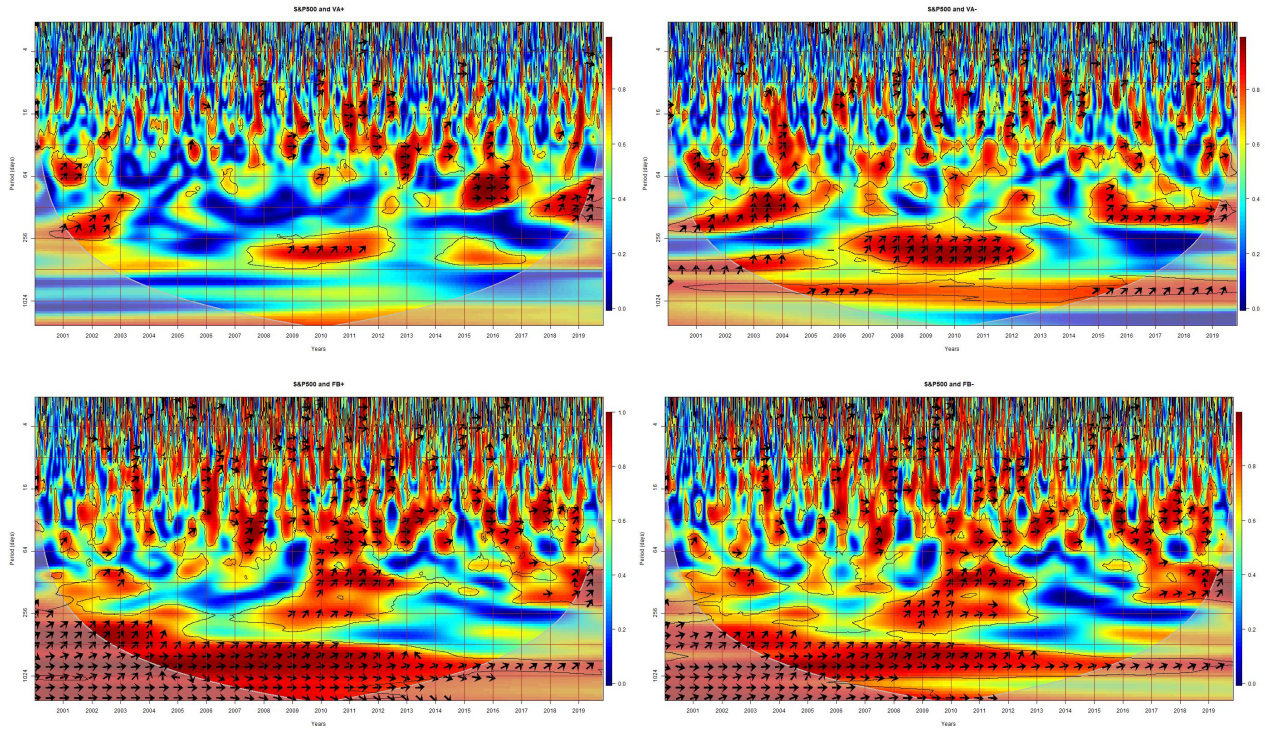


Figure 19: Full KE IRF plots.

A.3 Wavelet Plots

Wavelet plots for VA, FB, TR, KE, are shown below. Plot shows the coherence between the % returns of the S&P 500 and positive sentiment (left), and negative sentiment (right).



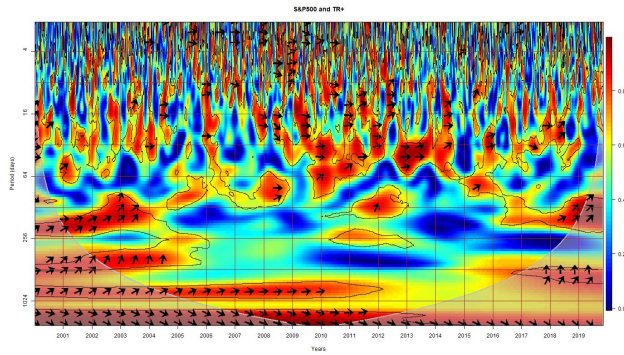


Figure 20: Wavelet coherence between the S&P 500 and positive news sentiments. From top to bottom: VA, FB, TR, KE.

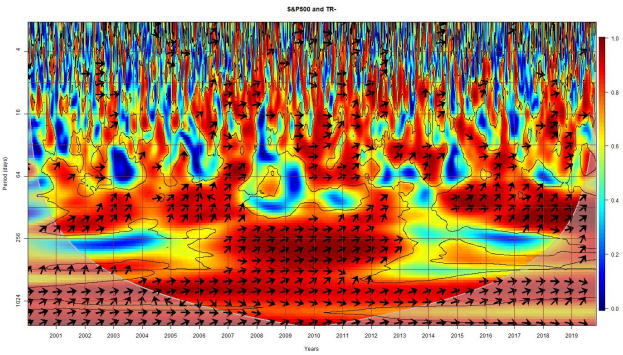
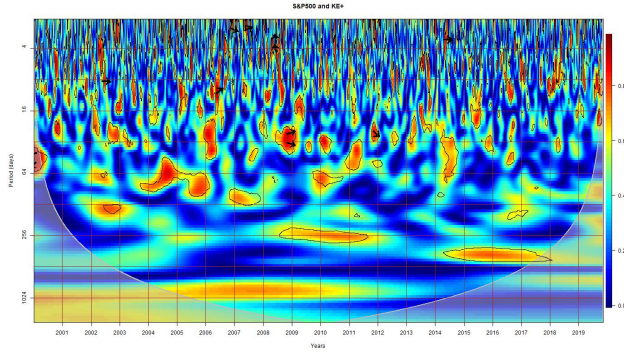
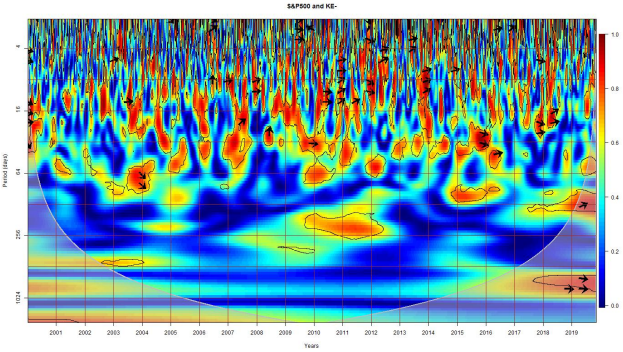


Figure 21: Wavelet coherence between the S&P 500 and negative news sentiments. From top to bottom: VA, FB, TR, KE.





B Appendix

B.1 LM Negate Keywords

The following negation keywords were used to change an otherwise positive word count to a negative word count for the Loughran & McDonald Dictionary classifier, LM:

aint, arent, cannot, cant, couldnt, darent, didnt, doesnt, ain't, aren't, can't, couldn't, daren't, didn't, doesn't, dont, hadnt, hasnt, havent, isnt, mightnt, mustnt, neither, don't, hadn't, hasn't, haven't, isn't, mightn't, mustn't, neednt, needn't, never, none, nope, nor, not, nothing, nowhere, oughtnt, shant, shouldnt, wasnt, werent, oughtn't, shan't, shouldn't, wasn't, weren't, without, wont, wouldnt, won't, wouldn't, rarely, seldom, despite, no, nobody.